

INTELLIGENT MULTISENSING DRONE DETECTION FOR UAV SURVEILLANCE

A Major Project Report Submitted in partial fulfilment of the requirements for the award of
the degree of

BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY

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Major Project - Phase - II



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MAY, 2026

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CERTIFICATE

This is to certify that the project report entitled “**INTELLIGENT MULTI-SENSING DRONE DETECTION FOR UAV SURVEILLANCE**” is a bonafide work done under our supervision and is being submitted by **Mr. M .Nikhil (22071A12G4), Miss. N. Rithwika (22071A12G6), Miss. N. Tejashwini (22071A12G9), Mr. N. Sai Mahesh (22071A12H0)** in partial fulfilment for the award of the degree of **Bachelor of Technology** in Information Technology, of the VNRVJIET, Hyderabad during the academic year 2025-2026. Certified further that to the best of our knowledge the work presented in this thesis has not been submitted to any other University or Institute for the award of any Degree or Diploma.

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DECLARATION

We declare that the major project work entitled “**INTELLIGENT MULTISENSING DRONE DETECTION FOR UAV SURVEILLANCE**” submitted in the department of Information Technology, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology** in **Information Technology** is a bonafide record of our own work carried out under the supervision of **Dr. G. Madhu Professor, Department of IT**. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

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ACKNOWLEDGEMENT

We express our deep sense of gratitude to our beloved **President, Sri. D. Suresh Babu**, VNR Vignana Jyothi Institute of Engineering & Technology for the valuable guidance and for permitting us to carry out this project.

With immense pleasure, we record our deep sense of gratitude to our beloved **Principal, Dr. B Chennakesava Rao**, for permitting us to carry out this project.

We express our deep sense of gratitude to our beloved Professor **Dr. N. Mangathayaru, Professor and Head of the Department**, Information Technology, VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad-500090 for the valuable guidance and suggestions, keen interest and through encouragement extended throughout the period of project work.

We take immense pleasure to express our deep sense of gratitude to our beloved Guides, **Dr. G. Madhu, Professor** in Information Technology, VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad, for their valuable suggestions and rare insights, for constant source of encouragement and inspiration throughout my project work.

We express our thanks to all those who contributed for the successful completion of our project work.

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ABSTRACT

Unmanned aerial vehicles (UAVs), also known as drones, pose potential threats in locations such as airports, military bases, and public spaces. Nevertheless, it is not easy to detect drones under challenging scenarios like low illumination and high environmental noise. The existing techniques used in drone detection are not accurate because they use a single type of sensing technique such as camera images, infrared, and audio signals.

This study aims at overcoming this challenge through proposing an advanced technique for the detection of drones based on multimodal information. It uses three types of information sources namely; visual information from RGB video frames, thermal information from infrared data, and finally auditory information from audio signals. Deep neural networks are utilized to learn spatial-temporal features from RGB and infrared video streams. Temporal features can be obtained from sequence input videos by employing suitable methods. Audio input signals are transformed into spectrogram form so as to recognize unique audio patterns of UAVs.

The benefit of such an approach is that, besides the solution being able to work effectively despite changes in the environment, detection precision will also improve. In fact, this approach has been realized as a software program on a computer, making it a valuable tool for detecting drones.

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LIST OF ABBREVIATIONS

S.No	Abbreviation	Meaning
1	CNN	Convolutional neural network
2	YOLO	You only look once
3	GRU	Gradient Recurrent unit
4	UAV	Unmanned Aerial vehicles
5	RCNN	Recurrent convolutional neural network
6	LSTM	Long Short Term Memory
7	UML	Unified Modelling Language

CHAPTER 1

INTRODUCTION

1.1 Increase in the Drone usage

The increasing availability of compact unmanned aerial vehicles has brought distinct benefits to commercial and recreational activities in recent years, while their growing presence in the environs of restricted zones has simultaneously introduced a wide range of security concerns. Encounters with unauthorized drones near areas such as airports, defense facilities, energy infrastructure, and crowded public spaces can cause disruptions to regular operations and raise both safety and privacy issues.

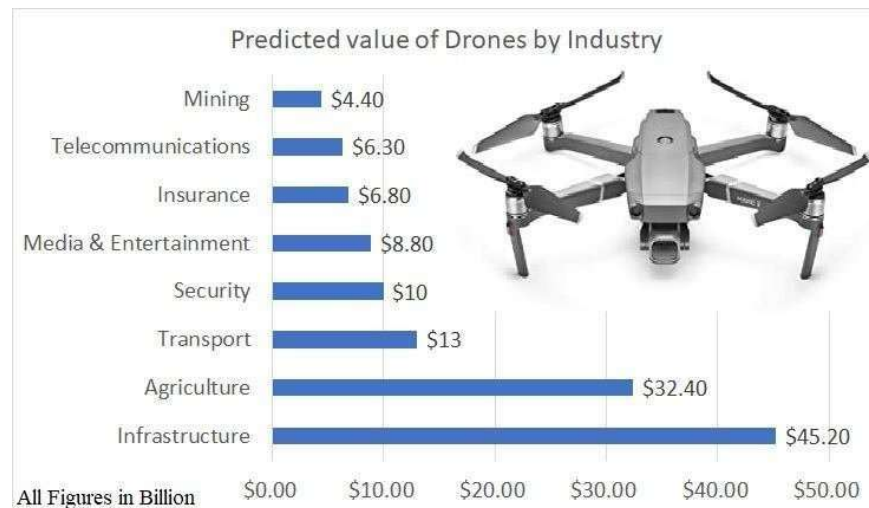


Fig 1.1 – Increase in the drone usage

The Fig 1.1 shows how the drones are a source of big business in most sectors, including mining and transport, media and security. The most likely sectors to experience high

investment are infrastructures and agricultural sectors and its projected amount is well over 30 and 40 billion dollars.

Even such industries as telecommunication and insurance will represent drones in the mass, which shows the growing popularity of this technology at the contemporary economy.

1.2 Background and Motivation

The drone has rapidly developed during the last ten years as a highly specialized military apparatus became affordable and could find applications in photography, delivery services, as part of industrial inspection, agriculture, and recreation.

The proliferation of them has brought about new opportunities- but new challenges also. Illegal drones have emerged around airports, government installations, civic events, and homes posing safety risk, privacy infringement, and a possible security risk. The small, low-flying UAVs do not actually pose a technological gap because the more traditional airspace monitoring technologies including radar were not initially built to track them.

The rising threats of security possession posed by drones have necessitated the use of detectors in contemporary settings. Drones may be exploited to spy, illegally conduct surveillance, traffic drug, or even cause malicious attacks thereby, posing great weaknesses to people, organizations, and the critical infrastructures. With a dependable detection system, situation awareness will be improved with an ability to detect early enough to prevent and reduce such incidents.

The safety in the airspace has also emerged as a big issue whereby Airports, stadiums and large-scale gatherings have been affected by unauthorized sightings of drones as illustrated in Fig.1.2. These cases underscore the importance of having effective surveillance strategies that could be used to identify and track UAVs until they become involved in disrupting the flight operations or crowd events to allow the safe and controlled use of the airspace.



Fig – 1.2 – Aircrafts usage expectations over the years

The quick pace of development of UAV applications in the sphere of commerce and also highlights how important the systematic regulation of the airspace should be. Monitoring tools need to develop, as drones are becoming more accessible and popular to use,

1.3 Emerging Detection Technologies

To deal with this, studies have narrowed down to new form of detection, including RF signal, acoustic, computer-vision and machine-learning-based classification. The

accuracy and strength in various detection techniques can be enhanced by a combination of two or more detection methods particularly in a complex environment.

More unmanned aerial vehicles enhanced the sovereignty of the state of art in detecting and classifying drones in various applications. With the increasing growth of the various UAVs development as shown in Fig 1.2 , the focus of researchers has shifted to effective real-time recognitions that can operate under viable conditions in a wide range of operating environment. It has not been left behind in terms of creative sensor technology, pattern analysis and classification structures that have facilitated security and a check of the airspaces [\[1\]](#).

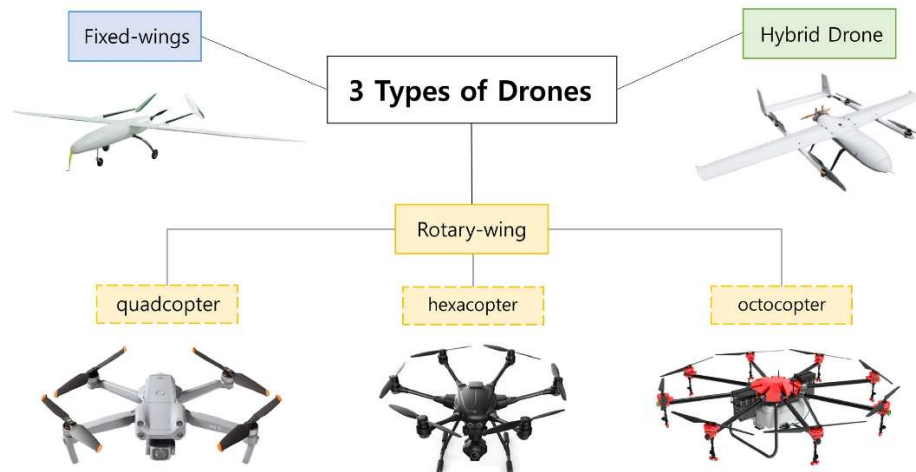


Fig 1.3 Overview of UAV Categories and Their Variants

The most significant achievement of these developments is the creation of specific datasets that reflect the issues of the long range detection. The accessibly large sized data

sets which resembles the real world conditions and types of operations allows both multi-modal and visual modes of detection to be tested equally and stringently even the new methods are tested on realistic operations conditions [\[2\]](#).

Fig. 1.3. demonstrates that there are varieties of drones classified under the key categories of fixed-wing, rotary-wing, and a hybrid category depending on their flying mechanisms and the organization of their internal structure. The requirement to fly over a distance over a long period is appropriate in fixed-wing drones because the drone is capable of gliding at distances. Quadcopters, hexacopters, and octocopters, known as rotary-wing drones are commonly used in monitoring and inspection works due to their hovering feature and ability to approach a complex environment.

Hybrid drones combine the benefits of a fixed-wing platform with that of a rotary-wing platform, allowing them to take off and land vertically, and fly forward efficiently. It is because of this difference of the drone structures, sizes, movement patterns and sensing signatures that it is a difficult task to detect drones in actual surveillance scenarios. As such, a powerful detection mechanism that will be capable of detecting different types of drones in diverse environmental factors is a significant design imperative of current intelligent UAV surveillance systems.

It also implied that massive scaled datasets of aerial and ground objects annotated are to be scaled since the quadcopters, as well as other drones, are becoming quite common. The resources have played an important role in making object detection algorithms very fine and testing and adoption of the algorithms over experimental systems to real time surveillance and monitoring review systems as a part of surveillance and monitoring.

Integration of multi-modes of data especially a combination of infrared and normal visual imagery to detect drones has been found to be particularly effective in further streamlining drone detection. The experiments carried out utilizing greater than one category of sensors defined the potential unique advantages that each of these modalities may produce on the sensor fusion approaches that are capable of enhancing the accessibility and strength of current and future operational environments [\[4\]](#).

The rapid changes in the world of counter-UAVs brought by these new developments in deep learning and deep learning using novel technologies to combine the data obtained by electro-optical, thermal, acoustic, radar, and RF devices in specific have changed the world of counter-UAVs in a speedy manner. According to surveys, there is the possibility of a combination of advanced neural network system and these multiple streams of data to take advantage of the optimum output of the detection and the interventions that can be implemented in the present to utilize the interventions in real-time [\[5\]](#).

Acoustic and RF have also stated that the methodologies need conventional measures of consideration. Lack of standard benchmarks is one of the outstanding concerns among the researchers who are trying to compare, integrate, and further enhancement of these technologies. is understandable that the solutions will meet the requirements of the present functioning and future outlook of the UAV technology in strenuous conditions as they keep advancing [\[6\]](#).

Recognition-based efficient datasets based on quadcopter recognition are becoming increasingly popular in practice, and possibly less convincing than the neural models may be, whether the validity of their effectiveness. It can be instigable to the comparison of

solutions that may guarantee scalability and performance stability, most notably, when the advanced security system is used at the big scale [5].

In addition to it, there has been significant improvement of it through the new deep learning algorithms including the implementation of the bidirectional LSTM networks that are trained through the assistance of the Bayesian techniques. All these sophisticated architectures coupled with the pre-management of the information has made it to locate drones effectively even within the variable light, weather and background conditions [7]. Another significant enhancement allowed to the video-based detection method is the further developed use of capsule networks and attention systems which allowed the detection of drones with a significantly greater accuracy including the cases when they were small, partially obscured or accelerated in high velocity through highly dynamic conditions [8].

1.4 Research Gaps

However, as much as there has been much improvement in the detection technology employed in drones, there are a number of significant gaps that have not been filled. Conventional airspace surveillance devices like radar are not best suited to track small, low-altitude UAVs that can have low radar cross-sections and are readily lost behind the background. This capability poses a devastating requirement of special detection mechanisms that will specifically address the modern traits of drones.

In addition, The publicly available datasets of drone detection are still scarce, especially in diverse weather conditions, drone types, camera angles and long-range environments, which limits the creation and the validation of trusted machine-learning models.

The other type of gap is the real-time processing capabilities. Many of the models prove to be highly accurate when tested off-line, they do not satisfy the computational requirements of continuous real-time monitoring of the airspace. This brings a disparity between scholarly prototypes and implementable, reality systems.

1.5 Problem Statement

To develop a smart, multisensing drone detection system, which will detect drones under a wide variety of real-world scenarios, through a combination of RGB, infrared, audio information. The system uses RCapsNet to learn spatial relations among the parts of the drone and utilizes GRU to learn temporal dynamics, which enable the successful detection, and classification of the object even in bad conditions, i.e. occlusions, low-light, and noise among others.

1.6 Objective

The goal of this project is to develop and implement an efficient system for drone detection which incorporates multiple sources of information including RGB video, infrared video and sound. It tries to leverage the potential in processing each kind of data, learning significant characteristics related to drone visual, movement and auditory patterns and merging them using an optimal fusion framework. It is also intended to increase detection accuracy, reduce false alarms, and achieve robust and reliable detection in various real-world scenarios, such as low lighting, noisy and occluded environments.

1.7 Scope of Work

The research will focus on developing a multi-modal detection system for drones that will target RGB videos, infrared images, and acoustic recordings. The research will cover the processing of data, feature extraction, and creation of models through deep learning methods that help in learning the spatial, temporal, and acoustic features of drones. Additionally, the research will include the application of a fusion technique for integrating multiple modalities, and evaluating the performance of the proposed system based on various scenarios. The research is limited to software development and evaluation based on available datasets.

1.8 Novelty

The proposed model utilises Recurrent Capsule Network (RCapsNet) which effectively extracts the visual features of the drone explaining the part-whole associations of the drone (rotors, arms and various body segments). This method is robust to rotations, scaling and viewpoints unlike traditional convolutions that lose spatial information.

Gated Recurrent Unit (GRU) is used to model the temporal dynamics across frames. Thus, the model can learn the flight patterns of the drones better and thereby can predict in an accurate and efficient manner.

The framework introduced makes use of various sensors such as RGB, IR videos, and audio, and hence improves the detection accuracy. The use of multiple sensors ensures a very accurate process under difficult situations such as low illumination, obstructed view and background noise.

A novel frame selection process based on motion is used, and only relevant frames are used. This helps in the reduction of computation time and improving the detection accuracy.

A novel dual background subtraction approach that applies DualMOG2 algorithm is applied in IR videos, therefore reducing thermal noise and background changes. This enables an improved motion detection method in IR videos.

Confidence-based sensor fusion is proposed whereby the data is given a different weightage.

1.9 Thesis Organization

The thesis is structured into seven chapters.

Chapter 2 reviews previous research related to drone detection methods, such as vision-based detection, audio-based detection, multimodal detection, and deep learning.

Chapter 3 discusses the proposed approach in detail, including the dataset, preprocessing steps, model architectures, and multimodal fusion methods.

Chapter 4 elaborates on the system's implementation process, which includes pre-processing stages, the workflow of the models, and system diagrams.

Chapter 5 discusses the results of the experiments conducted using the proposed system with various modalities and fusion techniques.

Chapter 6 concludes the study, providing the findings and contributions.

Chapter 7 provides future directions for improving the proposed system.

CHAPTER 2

LITERATURE SURVEY

Kabir et al. [9] presented a drone detection model that is based on deep learning and uses the YOLOv7 architecture to detect UAVs in multifaceted video streams. The authors highlighted that even the latest CNN-based detectors, including YOLOv8, YOLO-NAS and Faster-RCNN struggle with the issues of stability and accuracy, especially with training on mixed datasets of drones and birds. They have combined their methodology with prediction encompassing most of the steps of resizing, normalization, and data augmentation then trained on a mixed dataset with 57, 311 images of aerial drones and background birds.

The backbone of YOLOv7, their re-parameterized convolutional modules optimizing the gradient flow and served to improve the capabilities of the model to locate small UAV targets in various scenes. On experimental assessments, the YOLOv7 model is found to have a precision of 0.9509, a recall of 0.9656 and a map 0.5 of 0.9673, which is much higher than those of YOLOv8 and YOLO-NAS which have significantly higher training curves and inference reliability. The authors have shown that YOLOv7 is better in real-time working, and it is capable of generalizing to untested samples of YouTube video.

Samadzadegan et al. [10] created a system based on vision and distinguishes between two different types of drones and birds with the help of a deep convolutional neural network. Their works point out that sensors, such as radar, LiDAR, or thermal cameras are a common method of drone monitoring, but they tend to pose problems, such as high price, weight addition, or space obfuscation.

A collection of 10,000 images of the different types of multirotor, helicopters, and birds was gathered and divided into training and evaluation parts.. It was tested that the system achieved a total accuracy of 83.0, mAP of 84.0 and a score of 81 at the IoU. Findings of the experiments with the confusion-matrix shows that the method can identify the multirotors, helicopters, and birds in most instances even when they are found in the distant environment and in congested space.

Svanstrom et al. [11] Introduce a live drone positioning and tracking system developed on the integration of a number of sensorial inputs. The authors understand that none of the sensors is very strong to identify the UAVs that are very small and operate in every possible environment and condition and therefore combine the thermal infrared images, video on the visible range, the sounds and the wide-angle lens monitoring with ADS-B and GPS information.

They have a design that can resolve high volumes of airspace and retain high-resolution analysis of the small flying objects yet they are able to do so by combining wide-FOV observation and a steerable pan-tilt vehicle that aims the thermal and visible cameras at possible objects of interest. The established multi-sensor multi-target data (more than 203,000 annotated frames and various types of targets such as drones, birds, airplanes, and helicopters) were a great contribution of this work as they have become one of the most diverse and publicly available datasets in the study of drones.

The article also investigates detector performance variation with the increase of sensor-target distance. It is experimentally demonstrated that both infrared and visible sensors, trained under YOLOv2 architectures, have a balanced and similar level of precision,

regardless of the failure of the thermal sensor to get high spatial resolution. Notably, multi-sensor fusion decreases the number of false alarms in sensor outputs and enhances their overall reliability.

Tegler et al. [12] present a multimodal drone-detection system which is based on both audio sensing and visual input to approximate the location of the surrounding UAVs. The installation consists of twelve microphone arrays, a ring of nine fixed wide-FOV cameras and a high-resolution unit mounted on a mobile. Although direction-of-arrival information derived using multi-channel recordings combines both sound and vision, direction-of-arrival information is specifically highlighted in the study.

The authors demonstrate the ability to track motion along various flight paths using based on time difference of arrival measurements. The data set used in the work consists of the thirteen actual outdoor videos on which three types of UAVs were used: two quadcopters and a fixed wing drone recorded under different distances and directions. Human auditory listeners also filled the task of pointing to the source of sounds to compare the results, and the indicated results on the suggest that the proposed method is always better in comparison to human senses especially on the situation at shorter distances (less than 150 meters) or shorter distances.

Further experiments, using larger UAVs demonstrate excellent performance of multirotor platforms, with fixed wing aircraft becoming difficult to operate because of the non-uniform acoustic response and extensive silent glides. On the whole, the authors show that a well-thought audio pipeline can serve as a highly useful pre-preparation step, directing a camera to the target in order to follow it and classify it even more.

Wang et al and his company developed a good idea of scan and trace drones directly on edge appliances with vision cameras. They were able to address difficult problems such as finding small objects and retaining objects with a fast speed with low power. Their method is known as YOLO-DroneDet, which begins with YOLOv8 and introduces a multiscale dense connection architecture, which helps to transmit features effectively, particularly to objects of small size that are difficult to track down. It further employs a multilayer shared head to reduce the additional parameters and simplify the detection component.

To put it on edge gear, they pruned the model and employed knowledge distillation to reduce it without much accuracy loss. It consists of a dynamic camera on a gimbal that can change the view and is useful in tracking the camera dynamically in real time. items like trees or buildings. All in all, it provides a clean and quick vision system to drone spotting and following, specifically in tight edges, and introduces new concepts to the architecture and the method of implementation [13].

Khan and others et al have compiled a comprehensive description of the detection of drones with an emphasis on potential improvement needed to support no-fly zones as more and more ordinary people utilize drones. The article categorizes the tech as sound based, radio frequency, radar and camera, and explains what works and what does not such as the way noises can cause audio to be distorted or why visuals need a clear line of sight. The review advocates the integration of sensors and takes a keen interest in the manner in which machine learning has advanced each category individually. Sound systems detect sounds that the drone produces out of control, RF detects them using

control signals, radar interrogations micro-Doppler effects, and vision deep learning on images to locate them. Nevertheless, in the article, there are lamentations about areas of weaknesses regarding the range of detection, its maintenance in adverse weather or light conditions, and integrating it with the setup of real-time edges. It provides practical considerations to inform the next stage towards arrangements that are more reliable, inexpensive to operate and conform to regulations [14].

Table 2 – Literature survey

S No.	Title	Year of Publication	Authors	Algorithms& Techniques	Advantages and analysis
1	Drone Detection from Video Streams Using Image Processing Techniques and YOLOv7.	2024	Muhammad K. Kabir et al	1.YoloV7 Image 2.Preprocessing 2.Data agumentation	This paper demonstrates YOLOv7 based detector trained on a mixed drone–bird dataset can achieve stable precision
2	Detection and Recognition of Drones Based on a Deep Convolutional Neural Network	2022	Samadzadegan F et al	1. CNN vision detector 2. single sensor Input 3.multiclass recognition	Proves that a CNN can be reliably used to distinguish the drones, helicopters and birds. With set of 10,000 images.

	Using Visible Imagery.				
3	A Dataset for Multi-Sensor Drone Detection. Year of Publication	2021	Svanström F et al	1. Multi sensor fusion (ir, audio, rgb) video. 2. YOLOv2 based detectors	Provides an example of system that shows how the visible, thermal, audio, ADS-B, and GPS able to work in combination can greatly identify drones reliably when the distance is very long and false alarms are reduced to the minimum level.
4	Detection and Localization of Drones and UAVs Using Sound and Vision.	2025	Tegler E et al	1.TDOA and beamforming 2. Audio Pipeline To high resolution	Shows that camera guidance controlled by audio is more successful to localize and track drones in comparison with a human listener.
5	Drone Detection and Tracking: An Edge	2025	Wang B et al	1.YOLO Drone dataset based on YOLOv8 2.Vision	YOLO Drone Detection a lightweight and reduced variant of YOLOv8 that can track small and fast

	Deployable Efficient Algorithm Based on Vision Sensor.			sensor(dynamic tracking analysis)	drones in real-time.
6	On the Detection of Unauthorized Drones: Techniques and Future Perspectives	2022	Khan MA et al	1.Survey on sound based , rf radar 2.Focus on sensor fusion concepts	Vaguely defines state of the art of acoustic, RF, radar, and vision based drone detection, the benefits of machine learning in each of the modalities and how the fusion of multiple sensors is the key to overcoming the range, weather, and line of sight limitation.

7	An Acoustic Based Surveillance System for Amateur Drones Detection and Localization.	2020	Shi Z et al	1.Tetrahedral microphone clusters 2. SVM based per-array detection	Shows that an acoustic array system operating at high detection rates entirely using noise flighted spaces can detect them with a reported detectability of more than 90 percent in no-noise testing.
8	A lightweight low altitude drone detection model.	2025	Yu W et al	1.YOLO style backbone 2.Convolution blocks	Shows the ability of YOLO-GCOF to operate error performance with small drones.

Shi et al. implemented a real-time acoustic surveillance system that covers amateur drone surveillance and localization, based on the use of matrixes of microphone cluster tetrahedrons to provide 24/7 surveillance. Support vector machines are used to categorize the presence of drones per microphone and weighted fusion algorithm combines the results of individual microphones to enhance the detection strength. The localization uses time-difference-of-arrival (TDOA) calculated based on generalized cross-correlation with phase transfer, a new Bayesian filter-based algorithm is used.

Tests by evaluating the capabilities of the system to be used to detect drones at different distances as long as it was highly available confirmed the feasibility of the system in terms of operations to constantly monitor civilian drones despite the challenges posed by noise in the environment [15].

Yu and Mo et al propose YOLO-GCOF as a more realistic approach to obtain small drones that fly near the ground, as clutter, buildings, and trees typically pose a problem with traditional detectors. It considers the backbone of YOLO, though, instead of using heavier layers, the model uses lightweight convolution blocks and a slim feature-fusion neck to ensure that it can run on edge devices without the need to have a powerful GPU. Meanwhile, it introduces attention and loss-function modifications, which assist the network to concentrate on the finer aspects, such that small drones yet become conspicuous among the scene. In general, the results in the work indicate the possibility of maintaining high detection rate of low-altitude UAVs and reducing the computation and model size to enable the feasibility of continuous and real time monitoring in real use of the system [16].

Khan and co-authors et al create a system of the Internet of Things use to provide safety, where illegitimate or even suspicious drones are identified as a problem within a greater dilemma of smart-city security and not as drone-like cases. Their architecture will connect multiple edge devices and sensors, and an assembly of machine-learning models are then put together in a fashion that no single classifier has the final word on an object being a drone and not just an activity that will do no harm. The work also addresses practical considerations such as the overhead of communication, the cost involved in the

deployment and the capability of the architecture to be scaled as the community increases the number of cameras or sensors and otherwise, which are also used to make the architecture an attractive blueprint of reality to the solutions of community-level drone defense [17].

Rani R, Singh S, Choudhary S, Kumar A et al has adapted YOLOv5 to the specific needs of a drone by redesigning not only the feature extraction but also the interpretation process. The authors use GhostNet instead of the traditional backbone, allowing the network to be significantly lighter and speedy, this is significant when the surveillance requires a persistent and immediate execution on a minimal hardware. They also feed it with an Adaptive fourier Neural Operator (AFNO-2D) module which assists the model in capturing broader, global trends in the scene, thereby assisting it more easily in finding small or faraway drones that would otherwise be lost in the background. Combined, these changes provide a detector that is capable of being real-time, but even more fundamentally capable of identifying UAVs out of noisy, cluttered video [9].

2.1 Vision Based Drone Detection

The vision-based techniques of drone detection are widespread owing to the availability of optical cameras and efficiency of deep learning techniques for image processing. Typically, such methods involve using Convolutional Neural Networks (CNN) and real-time object detection algorithms to detect drones based on camera shots or video frames. Despite being highly accurate under the controlled conditions.

A number of research papers utilized YOLO networks for detecting UAVs in real-time mode. Thus, Zamri et al. developed an enhanced YOLOv8 network with attention

mechanisms, which was capable of identifying drones with mAP equaling 92.6 % at real-time detection speeds [18]. Nevertheless, prior research, which applied YOLOv4 and other CNN detectors, provided even higher accuracy values but failed to detect the target drones when they looked like birds and consisted of several pixels in the image [1][10]. Moreover, Zhao et al. introduced the DUT Anti-UAV dataset and showed that adding tracking to detection could significantly improve tracking accuracy (+20%) [19].

Table 2.1 – Visual Based Drone Detection

Work	Drone Type	Dataset	Pre-processing	Features	Detection Method	Results
[27]	Small UAVs, aerial targets	Aerial video sequences	Real-time frame processing	Learned features via attention modules	YOLOv8 + attention-based feature enhancement	mAP: 92.6% at real-time speed
[28]	Outdoor UAVs	DUT Anti-UAV dataset	Multi-frame sequence input	Learned spatial features	Faster-RCNN / Cascade-RCNN / YOLOX + multi-tracker fusion	Tracking success >20% improvement over single trackers
[14]	UAVs, varied backgrounds	UAV video sequences	Background subtraction preprocessing	Learned deep features	MobileNetV2 classification	Accuracy: 99.83%; degraded on non-stationary backgrounds
[17], [29]	Small UAVs, bird-like targets	Aerial image datasets	Standard CNN preprocessing	Learned convolutional features	YOLOv4 and CNN-based detectors	High accuracy; performance degraded on small or bird-like targets

Other methods as listed in Table 2.1 focus on improving system robustness via hybrid detection. Seidaliyeva et al. presented a two-step approach involving background subtraction and MobileNetV2-based classification, with an accuracy of 99.83%, but

dropping when multiple objects are in motion [20] However, vision-based approaches still suffer from environmental and quality factors such as light, camera resolution and long distances. Additionally, identifying drones over long ranges, due to the small size of the target, poses a challenge for distinguishing between a drone and a bird, thus requiring multimodal or fusion based solutions for effective real-world drone detection [4][8].

2.2 Audio Based Drone Detection

An alternative approach to visual drone detection is audio-based systems, which can be used in less-than-ideal conditions. The distinctive sounds generated by the rotating propellers and motors on drones can be used with signal processing and machine learning algorithms to classify them. Audio-based approach is cheaper than visual techniques and can be used at night, but it is sensitive to noise and distance to the source [4][14].

Table 2.2 – Audio Based Drone Detection

Work	Drone / Class	Description / Dataset	Model	Accuracy
[31]	Drone vs. environment	Audio recordings: Drone/Crowd, Drone/Nature, Drone/Train, Drone/Street	SVM (MFCC features)	99% (controlled); 94% (open-set)
[15]	Drone / Non-drone	UAV acoustic signals with background noise; spectrogram and STFT inputs	CNN + RNN	Improved over classical ML; sensitive to noise
[32]	Drone / Non-drone	Multi-environment audio; MFCC and frequency-domain inputs	Deep Neural Network	Competitive accuracy; limited in high-noise outdoor settings

Various research that is in Table 2.3 employ acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC) to classify drone sounds. Lee et al. used Support Vector Machine (SVM) classifiers, with almost 99% accuracy in noisy-free conditions and around 94% in noisy conditions, showing that MFCC-based methods are effective and efficient although they are less accurate in a more complex background [\[21\]](#).

Deep learning techniques like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) are used in audio-based drone detection to learn temporal and frequency-based features. Sonogram, Short-Time Fourier Transform (STFT), and MFCC features enhance classification accuracy over traditional approaches. But audio detection suffers from environmental noise, short-range, and has no spatial information about drone position, and needs to be complemented with other sensing technologies for better real-world accuracy.

2.3 MultiModal Drone Detection

Multimodal drone detection has emerged to address the limitations of single-sensor systems, which often fail under real-world conditions such as poor lighting, background noise, or signal interference. Recent studies combine data from multiple sensors including visible, thermal, acoustic, radar, and RF to improve detection accuracy and robustness. Sensor fusion enables one modality to compensate for the weaknesses of another, making multimodal approaches more reliable for real-time surveillance applications [\[8\]](#), [\[14\]](#). **Svanström et al.** proposed a confidence-based late fusion framework integrating visible, thermal, and audio sensors, where YOLOv2 was used for visual detection, LSTM for audio classification, and Kalman filtering for tracking. Their system achieved a detection

rate of 78%, compared to 67% for the best single sensor[18].

Multimodal architectures have also been explored in various sensing modalities as shown in Table 2.4. **Mandal and Satija** et al showed that combining time-domain and frequency-domain RF representations using a parallel CNN significantly improves classification performance over single-input models [22] Additionally, several review studies highlight that fusion-based systems using optical, thermal, acoustic, and radar data provide higher reliability in complex outdoor environments where drones appear as small and fast-moving targets [4][8].

Work	Drone Type	Modalities	Dataset	Features	Detection Method	Results
[23]	Outdoor UAVs	Visual, Thermal, Audio	Real outdoor UAV sequences	Learned visual, LSTM audio, Kalman tracking	YOLOv2 (visual) + LSTM (audio) + Kalman filter (tracking); late fusion	78% detection rate vs. 67% for best single-sensor
[24]	RF-emitting UAVs	RF signals (time)	frequency domain)	RF drone signal dataset	Time-domain and frequency-domain RF features	Parallel CNN with dual-input RF representations
[15], [32]	Various UAV types	Optical, Thermal, Acoustic, Radar	Complex outdoor datasets	Multi-sensor complementary features	Fusion-based deep learning systems	More reliable than unimodal in complex outdoor conditions

Table 2.3– Multi Modal Drone Detection

However, multimodal systems introduce challenges such as increased computational complexity, higher hardware cost, and the need for precise sensor synchronization. Despite these challenges, recent research indicates that multimodal deep learning frameworks are the most promising solution for accurate and robust drone detection in security-critical applications [14][18].

2.4 Deep Learning Based Drone Detection

The main approach to drone detection is based on deep learning models as they can bypass the feature engineering step while learning directly from large-scale datasets. CNN, RNN, LSTM, Capsule Networks are some of the models that have been applied to images, videos, audio, and radar for detection purposes which are summarized in Table 2.5. These methods offer superior accuracy over machine learning algorithms and allow detecting small sized drones moving at high speeds within complex backgrounds[4][8]. Object detectors like YOLO, Faster-RCNN, MobileNet that are based on Convolutional Neural Networks have shown promising results for real-time detection. Attention models and feature extractors that target multi-scale representations have also been introduced for small object detection. [18][20]

Table 2.4 – Deep Learning Drone Detection

Work	Drone Type	Dataset	Pre-processing	Features	Detection Method	Results
[27], [14]	Small UAVs, fast-moving targets	Aerial video datasets	Multi-scale feature extraction; real-time frame processing	Attention-based and multi-scale learned features	Improved YOLO networks; lightweight CNN	High detection rate; suitable for real-time scenarios
[23], [15]	UAVs in video / audio sequences	Multi-frame video; UAV audio signals	Temporal sequence preprocessing	Spatial and temporal features via CNN + recurrent layers	LSTM / GRU + CNN (joint spatial-temporal)	Better classification of motion and acoustic patterns
[25]	UAVs in high-resolution images	UAV aerial image dataset	Standard image resizing and normalization	Spatial hierarchy via dynamic routing (no pooling)	Capsule Networks	High detection accuracy with shallow architecture; preserves

						part-whole spatial relationships
[15], [32]	Various UAVs, complex environments	Mixed real-world datasets	Hybrid preprocessing pipelines	Complementary spatial, temporal, and acoustic features	Hybrid multimodal deep learning (CNN + RNN + advanced extractors)	Reliable performance across varied real-world conditions

Recurrent networks like LSTM and GRU models are used to capture temporal correlations from video frames and audio streams. [8][18] They can also be combined with CNN feature extractors for multimodal drone detection systems. Capsule Networks models are known to capture part-whole relationships between features and Mekhalfi et al. leveraged this property of Capsule Networks using dynamic routing to improve detection performance[7]. Due to the extensive data and computational power requirements of training deep learning models, multimodal and hybrid frameworks utilizing CNNs and recurrent models are preferred. [4][14]

CHAPTER 3

METHODOLOGY

3.1 Existing Systems

Various traditional methods and sensing technologies have been developed in this regard. Most of the existing solutions are based on a single type of input, such as radar signals, RGB video feeds, thermal images, or acoustic patterns, to identify the presence of a drone. Each of these approaches shows reasonably good performance for specific conditions; at the same time, they are also subjected to significant limitations while working in real-world environments. The variations in weather conditions, lighting, background complexity, and the small size of drones make accurate detection very challenging. Therefore, many current systems misidentify, result in inconsistent performance, and are less reliable.

One common method involves radar-based detection, whereby the drones are detected through analysis of reflected electromagnetic waves. Radar systems measure distance, velocity, and movement patterns; they often don't detect small drones because those offer weak reflections, especially in cases of flight close to the ground or in crowded urban spaces.

Another counter measure uses acoustic detection, based on picking up the unique sound made by drone propellers. While these systems can identify drone-specific audio signatures, performance drops significantly in noisy environments like cities, highways, and industrial zones, as the drone noise is masked by background sounds.

The widely used method today is RGB camera-based detection, whereby standard video cameras feed image frames into deep learning models like YOLO or Faster R-CNN. While these systems are able to detect drones very effectively during clear daylight, they suffer from major drawbacks in low-light, foggy, or overly bright conditions. They also tend to confuse drones with birds, leading to frequent false alarms.

Lastly, thermal or infrared imaging systems detect drones by capturing heat signatures. These systems perform well during nighttime or low-light situations but struggle with the low resolution of thermal images, which makes distinguishing small drones from warm background objects such as buildings, machinery, or even heat waves very difficult.

3.2 Issues and Challenges

The detection of drones in the natural environment is much more complicated than it may seem at first glance. First, there is a very close similarity between drones and birds. Both objects look small, dark, and of similar shape when viewed from a distance. Often, their flight behaviors also tend to overlap, or suddenly moving that cause detection models to confuse one for the other. This, in fact, increases false positives by a great degree and undermines the credibility of the existing systems.

Weather and lighting conditions are another important factor. Outdoor surveillance will obviously be interfered with by fog, rain, shadows, glare, or dust-all contributing to the distortion of camera inputs.

Another severe limitation is the small physical size of modern drones, especially consumer-grade micro-UAVs. In case they fly far from the observer, their images appear

as tiny dots, sometimes only a few pixels wide, making it extremely difficult for traditional deep learning models to extract meaningful visual features. Eventually, detection over a long distance is not reliable, and threats might not be noticed until it's too late.

Traditional sensors all have their shortcomings: radar systems cannot reliably capture small drones, Together, these limitations make it clear that relying on a single type of sensor isn't sufficient.

Another challenge is the insufficiency of diverse and high-quality training datasets. Most of the deep-learning-based models are trained on very limited data that may not cover all variations in altitude, background, weather, or even drone type. Without broad and well-balanced datasets, such models fail to generalize, leading to high error rates when tested outside controlled environments.

Another major concern involves the false positives. Birds, aircraft, balloons, clouds, and even insects can all trigger false detections. The difficulty increases further when the backgrounds are cluttered, like in skylines, forests, or crowded event spaces. Such false alarms may cause disturbances in operations and a loss of confidence in the system, especially within zones sensitive from a security point of view.

Moreover, most of the powerful detection models require high computational resources. The YOLO, SSD, and Faster R-CNN all require strong GPUs and large memory for smooth running; hence, real-time deployment on devices with limited processing power is hard to achieve. This further raises the cost while limiting scalability. There is also a bottleneck concerning real-time performance: fast-moving drones require faster

processing while maintaining track of the object; many models either lag, drop frames, or lose the drone upon sudden changes in direction. The tradeoff between accuracy and speed remains a significant hurdle.

Lastly, the erratic flight paths of small multi-rotors make detection even more difficult. These drones can accelerate quickly, make sharp rotations, and are prone to unexpected movements. Traditional detection methods usually fail in following such irregular motions, resulting in unstable bounding boxes or sometimes a complete loss of tracking.

3.3 Introduction to proposed methodology

The proposed framework works on a multimodal pipeline that independently handles RGB video, infrared video, and audio in separate preprocessing stages before merging their outputs through a decision-level fusion strategy. Thus, the methodology can be understood in three major parts: video preprocessing, audio preprocessing, and the final multimodal inference stage where fusion is done

3.4 Dataset Description

In this experiment, we employed the Zenodo Dataset that provides audio files, video(RGB) files, video(IR) files.[\[11\]](#) The dataset also contains matlab files for the video datatype. Each matlab file contains annotated frames of nearly 300 frames for each video(more than 203,000 annotated frames).

3.4.1 Audio Class : Dataset consists of 90 audio recordings of BACKGROUND, DRONE, HELICOPTER. And every recording is of length 10 seconds. The format of the files is .wav.

3.4.2 Video(RGB) Class : We are having 285 video recordings of AIRPLANE, BIRD, DRONE, HELICOPTER. The data captured with visible camera (IR cam)

3.4.3 Video(IR) Class : We are having 365 video recordings of AIRPLANE, BIRD, DRONE, HELICOPTER. The data captured with thermal infrared camera (V cam)

Table 3.1 –Total instances of each class in dataset

Class	Instances
Audio	90
Video(RGB)	285
Video (RGB) matlab files	285
Video(RGB)	365
Video (RGB) matlab files	365

Table 3.2 – Classes of each sensor

Sensor	Output Classes				
IR cam	Airplane	Bird	Drone	Helicopter	
V Cam	Airplane	Bird	Drone	Helicopter	
Audio			Drone	Helicopter	Background

3.5 Overview of Proposed Multi-Sensing System

In this section, the proposed multi-sensing based drone detection approach is discussed. The Fig. X demonstrates the proposed system architecture that applies to a combination of features extracted from three modalities of sensing (visible (RGB) video, infrared (IR) video and audio) for improving the accuracy and robustness of drone detection in complex environments.

The proposed system operates at three stages: data pre-processing, feature extraction and temporal modelling in each sensing modality and the multi-sensory fusion for final decision-making. In the preprocessing phase, the RGB and IR videos are converted into the fixed-length sequence of video frames and the audio signal is divided, and converted into log-mel spectrograms. Such preprocessed data are then fed into specific deep learning models that represent each amount of sensing.

When it comes to the vision part, we are able to identify the spatial information and running recurrent capsule neural network (RCapsNet) will be used to gain hierarchical spatial knowledge in objects and preserve them. The extracted spatial features of sequential frames are also given to a gated recurrent unit (GRU) network to get temporal features of the video. This helps the system to learn about object's spatial-temporal characteristics.

In the case of the audio signal, the spectral features of the audio are extracted using a convolutional neural network (CNN) that takes as input the log-mel spectrogram features. These features are then passed on to a GRU-based temporal model to pick up on the sequence in the audio signal to help the system distinguish between the sound of the drone and other audio artefacts.

The RGB, infrared and audio branches are transformed into probabilities denoting the likelihood of the occurrence of the drone. Probabilities of each of the modality are then combined, in what is called a late fusion mechanism, to yield the detection decision. Through such fusion, the system is able to exploit the complementary information of multiple sensors and when the performance of one or more modalities gets unreliable, this fusion mechanism results in better detection.

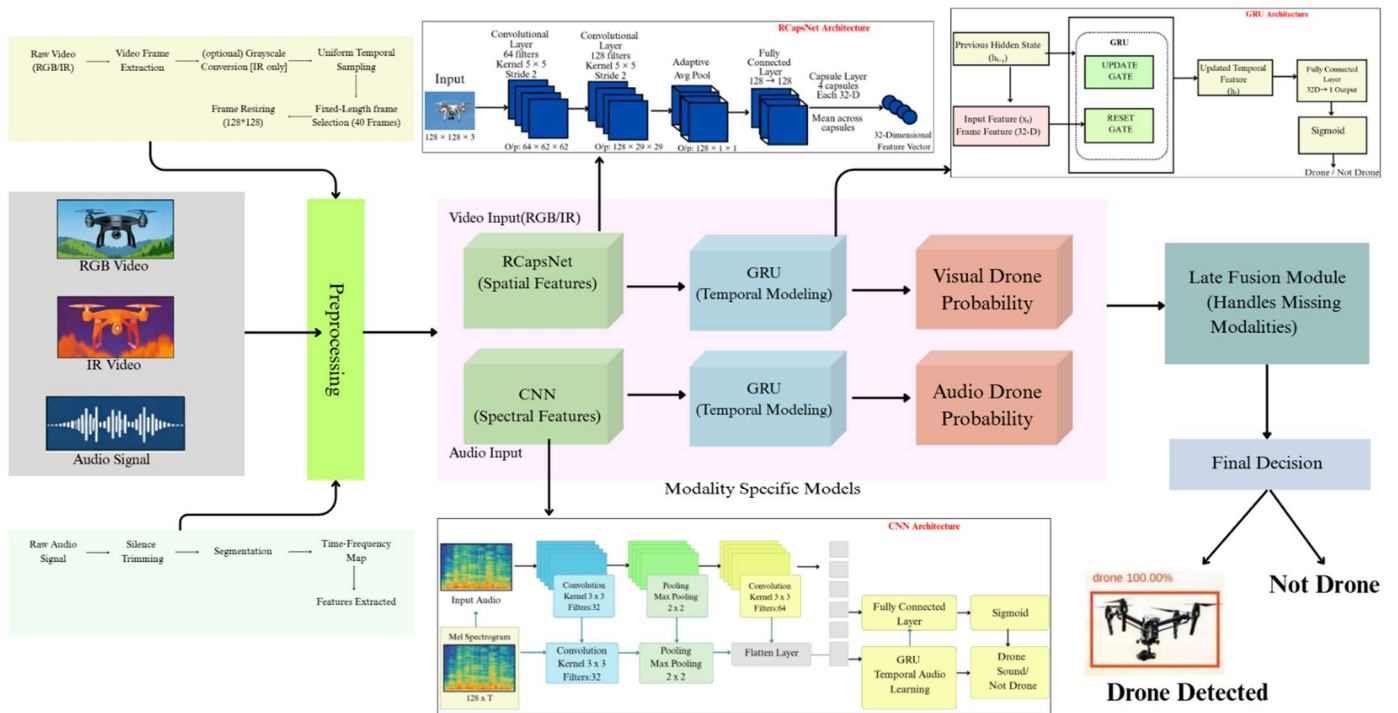


Fig 3. 1 – Architecture

3.6 Data Preprocessing and Feature Extraction

3.6.1 Training-Time Preprocessing

In the training stage, a standardized preprocessing pipeline is used to bring uniformity of all modalities, and to allow easy model learning.

In the case of RGB and infrared (IR) video data, each video sequence is first broken down into individual video frames. Forty equally sampled frames are taken over the duration of each video, to control time consistency. All the frames are downsampled to 128×128 pixels. In the case of RGB data, ImageNet mean and standard deviation are used to perform the standard normalization, which is followed by data augmentation schemes, namely, random horizontal flipping, slight rotation and color jittering to enhance the generalisation.

In the case of the IR data, a mean of 0.5 and a standard deviation of 0.5 are used to normalize the grayscale frames. In audio data, the raw signal waveforms (waveforms) are treated with a sliding-window method. All the audio files are divided into audio clips of 4 seconds with a 2-second overlap. A log-mel spectrogram (which uses 128 mel bands) is then calculated on each segment, then normalized and padded or trimmed down to a fixed-size representation of 128×247 . These spectrograms are input features of the audio classification model. This preprocessing on training time makes sure that all modalities are represented in a common and organized form to be fed into deep learning models.

3.6.2 Inference-Time Motion-Based Frame Selection

Unlike training phase, the inference stage uses a motion-aware preprocessing strategy, which can enhance the detection accuracy of the objects by ensuring that the attention is on informative regions of interest. This is termed as a gated inference pipeline, with only those frames that have important motion being fed through to additional processing.

In case of RGB video inputs a frame differencing (FrameDiff) method is employed to detect motion areas. Frames are sequentially turned into grayscale and improved with contrast-limited adaptive histogram equalization (CLAHE). The value of the absolute difference at each pair of successive frame is calculated and then thresholded and morphological processes are done to produce motion masks. Candidate frames are frames which contain valid motion contours. Based on these motion indices, we uniformly sample 40 frames to create the input sequence to the RGB model.

In the case of IR video inputs, a dual background subtraction (Mixture of Gaussians (DualMOG2)) method is used. To capture the changing motion, two background models of different rates of updating (slow and fast) are used. The resulting masks are amalgamated and trimmed with morphological filtering to get motion regions. Just like the RGB pipeline, the frames that represent detected motion are picked, and 40 frames are sampled to make model inferences.

Also, a tracking mechanism based on contour is used to track objects in motion in a frame. Bounding boxes formed around observed motion areas are further stabilized with a Kalman filter and temporally consistency ensured during the visualization stage. Like the training process, the same sliding window method during training is also applied during

inference in the case of audio inputs. An independent processing is done on each segment to create log-mel spectrograms and predicting at the segment level can be added together to give the final audio probability. This frame selection motion scheme plays a crucial role in minimizing the effect of irrelevant parasite frames and has a stronger likelihood of picking up significant temporal features in the model and improves detection resilience in the real-world context.

Although preprocessing guarantees organization in input representation, extraction of features is implicitly carried out in the deep learning models. The spatial features are trained in the RGB and IR pipelines using capsule and convolutional layers, and temporal relationships are achieved with the help of recurrent units. In the case of audio modality, the log-mel spectrograms have a time-frequency representation that is then fed through convolutional layers to obtain discriminative acoustic features.

3.7 RGB Drone Detection Model

3.7.1 RCapsNet Feature Extraction

Recurrent capsule-based detectors are aimed at detection drones of rich spatial features using video frames. The input videos are represented as a series of $T=40$ frames, and each frame is rescaled to $128 \times 128 \times 3$.

First, all frames are filtered by a convolutional feature extractor, which is composed of cascades of convolutional filters with nonlinear activation functions. These layers convert the input frame into a smaller feature representation because they encode low-level and mid-level visual features like edges, textures and object structures.

Let $x_t \in R^{128 \times 128 \times 3}$ denote the input frame at time step t. The convolutional layers produce a feature vector $f_t \in R^{128}$ which is further transformed using a fully connected layer:

$$h_t = W_t f_t + b_f$$

Where W_t and b_f represent the learnable parameters.

Capsule layer is added to the transformed features to represent hierarchical relationships between spatial features. The capsule layer encodes the input feature by resulting in several higher-dimensional capsules:

$$u_k = W_k h_t$$

In which W_k is a transformation matrix of the k-th capsule. Non-linear squashing of the output of each capsule is performed:

$$v_k = \frac{\|u_k\|}{1 + \|u_k\|^2}$$

This squashing step takes care of the fact that the probability of the appearance of a given feature is encoded in the length of the capsule vector, whereas instantiation parameters (pose and scale) are represented using its orientation. The result of the all capsules are summed to gain a small s-wide feature representation:

$$Z_k = \frac{1}{K} \sum_{k=1}^K v_k$$

where K is the number of capsules.

3.7.2 GRU Temporal Modeling

Whereas RCapsNet learns spatial features of single-frame data, the time-related connections between data frames are captured by a Gated Recurrent Unit (GRU). This enables the model to handle motion patterns and time consistency in drone trajectories. Considering the order of the spatial features: $\{z_1, z_2, \dots, z_T\}$, the GRU updates its hidden state as follows:

$$h_t = GRU(z_t, h_{t-1})$$

GRU is made of update and reset gates which regulate the flow of information:

$$r_t = \sigma(W_r z_t + U_r h_{t-1})$$

$$u_t = \sigma(W_u z_t + U_u h_{t-1})$$

$$\{\widehat{h}_t\} = \tanh(W_h z_t + U_h(r_t \odot h_{t-1}))$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \{\widehat{h}_t\}$$

with $\sigma(\cdot)$ defined to be the sigmoid function and $\odot$ the element-wise product. Once all the frames have been processed the final hidden state h_t represents a summary of how the input sequence varies with time.. To enhance generalization, a dropout layer is used and a fully connected layer afterwards is used to generate the output logit:

$$y = W_o h_T + b_o$$

The final probability of drone presence is obtained using a sigmoid activation:

$$P_{\{RGB\}} = \sigma(y)$$

where $P_{\{RGB\}} \in [0,1]$ represents the probability that the input RGB video contains a drone.

3.8 Infrared Drone Detection Model

3.8.1 RCapsNet Feature Extraction

The structure of the IR drone detection algorithm resembles that of the RGB one but is adjusted for single-channel infrared images. Each video in IR format is modeled as a series of $T=40$ gray-scale frames, with each frame normalized into $128 \times 128 \times 1$.

Unlike RGB videos, infrared images detect temperature differences, allowing for reliable detection in low light and dark settings. Nevertheless, they usually lack texture and colors, making it difficult to extract features. In this regard, we apply a recurrent capsule-based network for extracting spatial relations from thermal images.

Let $x_t \in R^{128 \times 128 \times 3}$ denote the IR frame at time step t . The initial processing is performed via a few convolutional layers, resulting in a vectorized feature extraction: f_t , which is then transformed using a fully connected layer:

$$h_t = W_f f_t + b_f$$

A capsule layer is then applied to model hierarchical spatial features:

$$u_k = W_k h_t$$

The capsule outputs are aggregated to obtain the spatial feature vector:

$$v_k = \frac{\|u_k\|}{1 + \|u_k\|^2}$$

This representation enables the model to retain important structural information of small thermal targets such as drones.

3.8.2 GRU Temporal Modeling

To model the dynamic properties of IR videos, the spatial features extracted are fed into a Gated Recurrent Unit (GRU), similar to the RGB model. Given the sequence $\{z_1, z_2, \dots, z_T\}$, the hidden state update of the GRU is given by:

$$h_t = GRU(z_t, h_{t-1})$$

In this way, the motion patterns and temporal dependencies of the thermal signatures can be captured using the GRU, which is especially critical for IR videos where the visual cues are not as distinct.

The final hidden state h_T is then fed into a dropout layer and then the fully connected layer whose output logits are given by:

$$y = W_o h_T + b_o$$

The probability of the drone being present in the IR sequence is given by the sigmoid function:

$$P_{\{IR\}} = \sigma(y)$$

where $P_{\{IR\}} \in [0, 1]$

3.9 Audio Drone Detection Model

3.9.1 Log-Mel Spectrogram Features

Audio data can be helpful in complementing video data in drone detection as it can contain information about acoustic characteristics produced by the drone's motor. Audio signals, however, cannot be processed as they are because they are unsuitable for training deep neural networks. The transformation of an audio waveform into its time–frequency representation is done with log-mel spectrograms.

An input signal $y(t)$ is first split into 4-second overlapping segments with an offset of 2 seconds between each. For each segment, the mel spectrogram is computed through a short-time Fourier transform (STFT):

$$S(f, t) = |STFT(y(t))|^2$$

The frequency axis is then transformed to the mel frequency scale through filter bank transformation that produces a mel spectrogram $M(m, t)$. Lastly, the following logarithmic transformation is applied:

$$X = \log(M(m, t) + \epsilon)$$

where ϵ is a small constant to avoid numerical instability.

The resulting log-mel spectrogram $X \in R^{128 \times 247}$ captures both temporal and spectral characteristics of the audio signal and serves as input to the deep learning model.

3.9.2 CNN Feature Extraction

To extract discriminative features from the log-mel spectrogram, a convolutional neural network (CNN) is employed. The CNN learns local time–frequency patterns such as harmonic structures and frequency modulations that are characteristic of drone sounds.

Given the input spectrogram X , convolutional layers apply filters to extract feature maps:

$$F = \sigma(W * X + b)$$

where $*$ denotes convolution, W represents learnable filters, b is the bias term, and $\sigma(\cdot)$ is a nonlinear activation function.

Multiple convolutional blocks with batch normalization and max-pooling are used to progressively reduce spatial dimensions and enhance high-level feature representations.

The output of the CNN is a feature tensor $F \in R^{C \times F \times t}$

where C denotes channels, and F , T represent frequency and time dimensions, respectively.

These features encode important acoustic patterns that distinguish drone sounds from background noise and other aerial objects.

3.9.3 GRU Temporal Modeling

For capturing the temporal relationship between the elements of the signal, the output features from the CNN layers are further used in a GRU model. This feature matrix is transformed to a sequence of feature vectors:

$$z_t \in \mathbb{R}^C$$

The GRU processes the sequence as:

$$h_t = GRU(z_t, h_{\{t-1\}})$$

The GRU model captures the temporal dynamics within the acoustic signals like rotor noises and frequencies that are key factors in detecting the drone

The final hidden state h_T is passed through a fully connected layer to obtain the output logits using the following formula::

$$y = W_o h_T + b_o$$

The softmax function generates the class probabilities:

$$P_{\{Audio\}} = softmax(y)$$

where $P_{\{Audio\}}$ represents the probability of belonging to each class (Drone/Not Drone)

3.10 Fusion Strategy for Multi-Sensor Detection

To improve the robustness and reliability of drone detection, the proposed system integrates information from RGB, infrared (IR), and audio modalities using a late fusion strategy. Each modality independently produces a probability score indicating the likelihood of drone presence, represented as $P_{\{RGB\}}$, $P_{\{ir\}}$, $P_{\{Audio\}}$ respectively. These outputs are combined to obtain a final decision.

Initially, a static weighted fusion mechanism is employed, where fixed weights are assigned to each modality based on their relative contribution. The final static probability is computed as a weighted average of the individual modality probabilities. In this work, higher weights are assigned to visual modalities (RGB and IR) due to their strong spatial representation capability, while a comparatively lower weight is assigned to the audio modality.

Another way to boost adaptation is by adopting dynamic fusion. Here, weights are dynamically determined according to confidence. Confidence of a modality is defined in terms of the distance of the probability value of a modality from the decision boundary (0.5). If a prediction is close to 0 or 1, then it is considered confident, and if it is close to 0.5 it is considered uncertain.

Based on this idea, a confidence value is calculated for each modality and incorporated to dynamically adjust the weights. The dynamically weighted fusion probability is then computed as a weighted average of the modalities based on their score. This dynamic approach enables the system to give preference to modalities with high confidence and down weight those with low confidence.

Lastly, a thresholding decision rule is used with the final probability. If the final probability is equal to or larger than 0.5, the object is deemed to be a drone; else a non-drone.

The proposed fusion approach fused multimodal sensors' information effectively to improve detection accuracy in complex environments where each sensor could be unreliable.

3.11 CASE STUDY:

Video Model (RGB/IR)

Step 1: The Raw Input

Assume 1 RGB frame resized to 3×3 (instead of 128×128).

We are looking at only one channel.

$$\text{Input Grid} = \begin{bmatrix} 10 & 200 & 30 \\ 50 & 150 & 20 \\ 10 & 10 & 10 \end{bmatrix}$$

Step 2: Normalization :

To Tensor : Divide by 255

- Pixel 200 becomes: $200/255 = 0.78$
- Pixel 50 becomes: $50/255 = 0.196$

Normalize using formula :

$$x_{\{norm\}} = (x - mean)/std$$

Using your RGB values:

$$Mean = 0.485, Std = 0.229$$

For that bright pixel (0.78):

$$(0.78 - 0.485)/0.229 = 1.29$$

For the dark pixel (0.19):

$$\frac{0.19 - 0.485}{0.229} = -1.28$$

$$\text{Normalized Grid} = \begin{bmatrix} -1.5 & 1.29 & -1.4 \\ -1.28 & 0.5 & -1.3 \\ -1.5 & -1.5 & -1.5 \end{bmatrix}$$

Step3 : Convolution:

Let's use a tiny 2x2 Kernel (filter) that looks for a vertical edge. :

$$\text{Kernal} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

We place this kernel over the top-left corner of our Normalized Grid

$$\text{patch1} = \begin{bmatrix} -1.5 & 1.29 \\ -1.28 & 0.5 \end{bmatrix}$$

$$\text{Math} = (-1.5 \times 1) + (-1.28 \times 1)$$

$$\text{result} = -1.5 - 1.28 = -2.78$$

$$\text{patch2} = \begin{bmatrix} 1.29 & -1.4 \\ 0.5 & -1.3 \end{bmatrix}$$

$$\text{Math} : (1.29 \times 1) + (0.5 \times 1)$$

$$\text{result} = 1.29 + 0.5 = 1.79$$

$$patch3 = \begin{bmatrix} -1.28 & 0.5 \\ -1.5 & -1.5 \end{bmatrix}$$

$$Math = (-1.28 \times 1) + (-1.5 \times 1)$$

$$result = -1.28 - 1.5 = -2.78$$

$$patch4 = \begin{bmatrix} 0.5 & -1.3 \\ -1.5 & -1.5 \end{bmatrix}$$

$$Match : (0.5 \times 1) + (-1.5 \times 1)$$

$$result = 0.5 - 1.5 = -1.0$$

Step 4 : Capsule Layer

The Vectors from 4 capsule will get averaged for each frame and the vectos for each frame will be take by GRU for temporal modeling.

$$S = [3.0, 4.0]$$

$$1. \text{ Calculate Magnitude : } ||S|| = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$$

$$\text{Squash Formula : } v_k = \frac{\|u_k\|}{1 + \|u_k\|^2}$$

$$part1 = 5^2 / (1 + 5^2) = 25 / 26 \approx 0.96$$

$$part2 = [3, 4] / 5 = [0.6, 0.8]$$

$$Final Output = 0.96 \times [0.6, 0.8] = [0.576, 0.768]$$

Audio Model :

Take a tiny slice of the spectrogram.

- Rows = Frequency (Pitch)
- Columns = Time

$$Raw\ Energy = \begin{bmatrix} 10 & 10 \\ 1000 & 1000 \end{bmatrix}$$

Top row: Low energy (silence/wind).

Bottom row: High energy (Drone Buzz).

Step 2 : Log – mel Conversion :

Normalizing to decibel values

Formula : $10 \times \log_{10}(value)$

$$CELL(1,1) : 10 \times \log_{10} 10 = 10\ dB$$

$$CELL(3,3) : 10 \times \log_{10} 1000 = 30\ dB$$

$$New\ Grid = \begin{bmatrix} 10 & 10 \\ 30 & 30 \end{bmatrix}$$

Step 3 : Max Pooling:

We look at our 2x2 grid and picks the single biggest number.

$$Pool = \begin{bmatrix} 10 & 10 \\ 30 & 30 \end{bmatrix} = 30$$

This reduces the data size while keeping the most important feature

Step 4 : GRU

The GRU has a "Hidden State" (h) that acts as memory.

$$H_{old} = 0.5$$

$$X_{new} = 0.8$$

Let weights $W=1, U=0.5$

$$\text{Calculation: } (1 * 0.8) + (0.5 * 0.5) = 1.05$$

$$\text{Apply Tanh: } \tanh(1.05) \sim 0.78$$

Result: The memory updated from 0.5 to 0.78

Fusion:

Scenario A : All sensors available

$$P_{audio} = 0.80$$

$$P_{rgb} = 0.40$$

$$P_{ir} = 0.90$$

1. The code multiplies each probability (P) by its importance (w)

and adds them up.

$$Sum = (0.8 \times 0.25) + (0.40 \times 0.40) + (0.9 \times 0.35) = 0.675$$

2. Since everyone is present, we add up all weights:

$$W_{\{total\}} = 0.25 + 0.40 + 0.35 = 1.0$$

The Prediction Since $0.675 \geq 0.5$

Result = Drone

Scenario B : During the absence of a sensor

$$P_{\{audio\}} = NO INPUT$$

$$P_{\{rgb\}} = 0.40$$

$$P_{\{ir\}} = 0.90$$

1. Multiplies each probability (P) by its importance (w) and adds them up.

$$Sum = (0.40 \times 0.40) + (0.90 \times 0.35) = 0.475$$

2. Add the weights of active sensors :

$$W_{\{total\}} = 0.40 + 0.35 = 0.75$$

$$P_{\{final\}} = \frac{Sum}{W_{\{total\}}} = \frac{0.475}{0.75} = 0.633$$

The prediction without normalization , the score is 0.475. because of normalization score is 0.633 results in DRONE.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

The proposed framework works on a multimodal pipeline that independently handles RGB video, infrared video, and audio in separate preprocessing stages before merging their outputs through a decision-level fusion strategy. Thus, the methodology can be understood in three major parts: video preprocessing, audio preprocessing, and the final multimodal inference stage where fusion is done.

4.2. Video Preprocessing Pipeline

4.2.1 Acquisition of Raw video and Frame extraction.

The suggested drone detection system starts with the raw video sequences of the RGB and infrared (IR) modalities. Every video is sampled identically in a sequence of $T = 40$ frames in order to not maintain temporal consistency across inputs. Standardization of the input dimensions to 128×128 pixel spatial resolution and reduction of computationally complicated frames are performed to extract such frames.

The suggested drone detection system starts with the raw video sequences of the RGB and infrared (IR) modalities. Every video is sampled identically in a sequence of $T = 40$ frames in order to not maintain temporal consistency across inputs. Standardization of the input dimensions to 128×128 pixel spatial resolution and reduction of computationally complicated frames are performed to extract such frames.

4.2.2. Frame normalization and resizing.

Once the frames have been extracted, all the frames are resized into the same constant resolution of 128 x 128 pixels. This is done to provide consistency throughout the dataset and to be able to process them in bulk during training and inference.

Also, pixel intensity values are normalized to enhance convergence in the model and stabilize training. Normalization process minimizes variations due to differences in illumination and sensor properties between RGB and thermal modalities of imaging.

4.2.3. Temporal Sequence Construction

The normalized and removed frames are grouped into temporal blocks (40 consecutive frames). This representation of the sequence form enables the model to model the continuity of motion and dynamic appearances appearance changes of drones over time. Temporal sequencing is significant to isolate drones and background clutter since drones have some unique motion patterns that are not efficiently implied with one-frame analysis.

4.3. Audio Preprocessing Pipeline

4.3.1. Audio Resampling and Standardization

First, the audio is resampled to sixteen kilohertz so that all recordings operate under the same resolution and sampling characteristics. It then undergoes a silence-removal step with a threshold of forty decibels, to remove long segments of background noise that would add little value to the analysis and could interfere with later stages of processing.

4.3.2 Cleaning(Trimming)

It involves removal of noise and silence.

Math : For loudness we use Logarithms(Decibels) to convert the raw amplitude(A) to decibels(dB).

$$dB = 20 * \log_{10} |A|$$

4.3.3. Segmentation

The audio is segmented into overlapping portions with the use of a four-second window and a hop of two seconds after normalization. This results in a sequence of slices that preserve continuity in acoustic patterns while still allowing enough variety for learning. Every slice is then mapped to its time–frequency representation by a feature extraction process that emphasizes how signal energy is spread across time and frequency.

window1 : 0s to 4s

window2 : 2s to 6s(50% overlap)

window3 : 4s to 8s

4.3.4 Feature Extraction

It performs a Discrete fourier transformer

$$X[K] = \sum_0^{N-1} x[n] \cdot e^{-i2\pi kn / N}$$

$x[n]$ = Raw audio numbers

$X[k]$ = Strength of the frequency “K”

4.3.5. Normalization and formatting of output

All time frequency representations are padded or trimmed to a fixed size (128,247) in order to keep the input dimensionality consistent for the neural network. The final features

are saved in a standardized array Format, allowing them to be fed directly into the convolutional and recurrent models used in the audio branch.

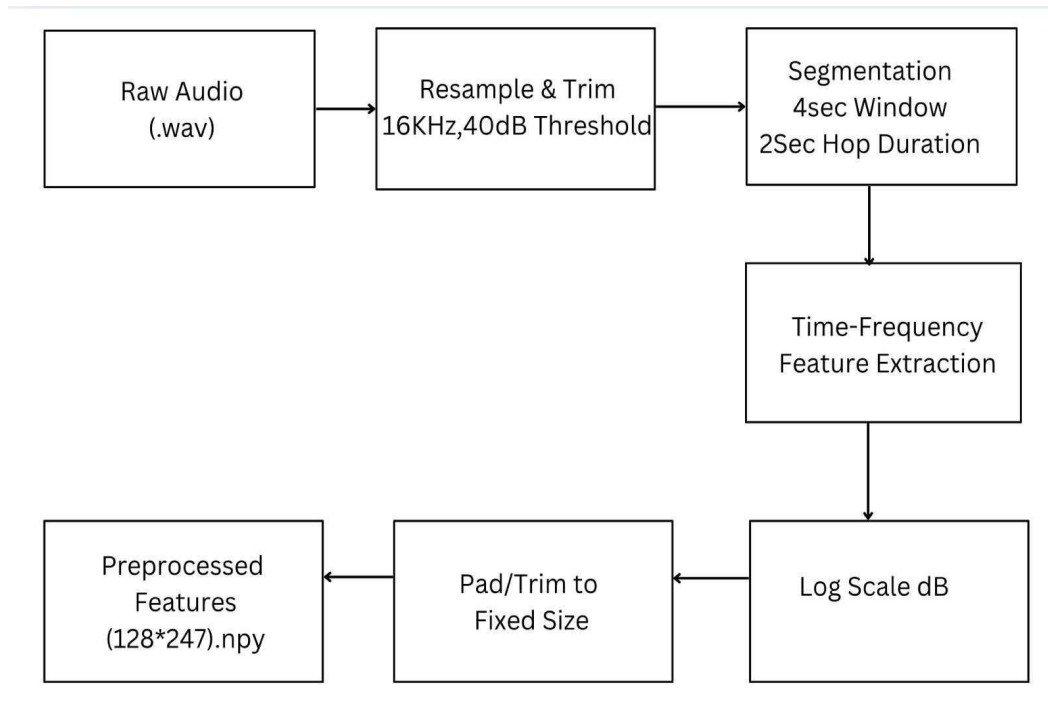


Fig 4.1 - Audio Preprocessing Architecture

The diagram Fig 4.2 represents Audio Preprocessing Architecture. Resample individual recordings, trim and cut them smaller and overlapping time-frequency images time-frequency images, which encode the changing nature of sound with time. Those views are translated into decibel and downsampled to average 128x247 format which are later saved as feature files that could easily be understood by a neural network.

4.4. Multi Model inference Pipeline :

4.4.1 . Independent Modality Processing

The input to this inference system can be a combination of RGB video, infrared video, or audio. The key idea is that each modality goes through its respective branch without having any information about the others. If there is an audio feed, it goes into the audio pipeline, yielding a confidence about the drone presence using that modality.

If both RGB and infrared inputs are given, then each will be processed exclusively by the visual pipeline, creating an independent visual source with a confidence score. Outputs are estimates about the likelihood of drone presence inferred by each of the modalities.

4.4.2. Capsule Network

Residual Capsule Network or RCapsNet is just a better version of the original concept of Capsule Network. Capsule Networks were supposed to identify patterns more intelligently as compared to normal CNNs, identifying not features themselves, but how they are related to other features. The RCapsNet introduces further improvement by addition of residual connections, which allows the network to learn more easily and prevents the problem of training that can be common with deep models. This practically means that the RCapsNet will be able to go deeper and learn more complex relations in images; in general, it will do better in induction tasks like classification or segmenting. It retains all the advantages of using capsule networks but makes them easier to learn to use and more stable.[\[19\]](#)

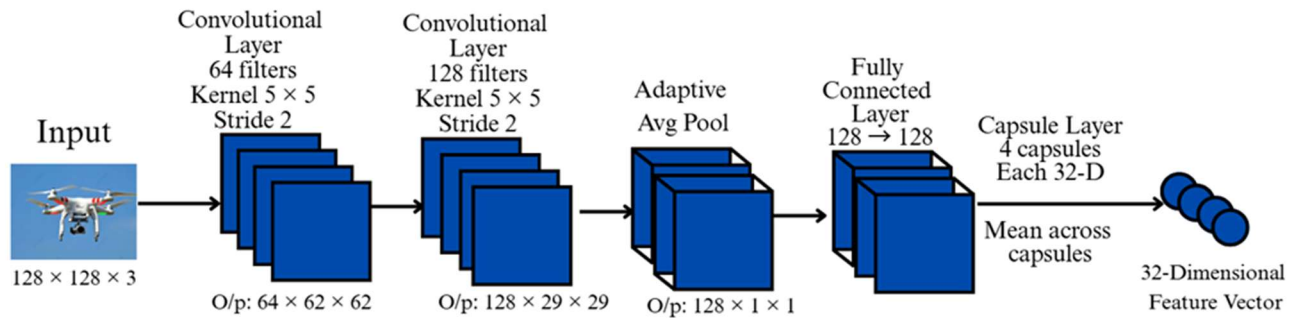


Fig 4.2 – RCapsNet Architecture

Fig 4.3 indicates that a capsule network silently makes a decision whether an image relayed contains a drone or not. Conventionally, a convolution layer extracts the low-level patterns of the picture and then the primary capsules cluster the patterns into more advanced components of the object.

Lastly, the class capsules and softmax layer take it all into consideration and delivers a mere verdict: is drone or is not-drone.

4.4.3. CNN

Convolutional neural network is a class of deep learning that works particularly well in the visual image perceptions field. A CNN sees it in parts unlike looking at a picture at once because we are more inclined to watch lines, color, and shape before we can focus on the entire object. These minute scans are termed convolutions and are utilized in determining the patterns within the network like corners, textures, or the entire object as you progressively further into the layer. Because they can learn on a layer-by-layer basis, CNNs can be very good at tasks involving image recognition, object detection, and some robotics tasks such as interpreting medical scans or driving a vision-based system.

Instead, they become among the most crucial components of modern AI in that they are learned automatically, and without a human providing them with any rules.

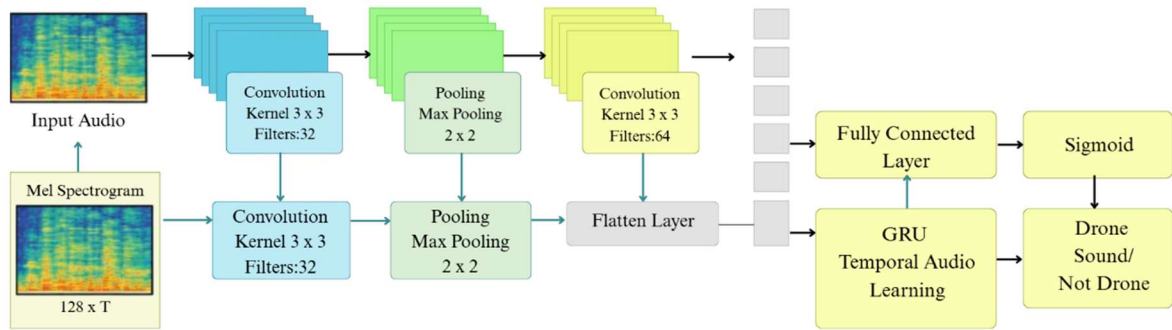


Fig 4.3 CNN architecture

4.4.4 GRU

GRU is a parameter sparse, area-friendly, and efficient recurrent Neural Network. A GRU manages to simplify the arrangement without the need to use a number of gates through the adoption of two: an update gate and a reset gate. These gates assist in decision-making to the model concerning what information should be recalled, what information should be forgotten, and the degree of the past information to influence the present output. GRUs are strong enough of learning faster in time-series data. They form part of the appealing alternatives in most practical uses where quickness and effectiveness is important since they can make similar performance as LSTM.

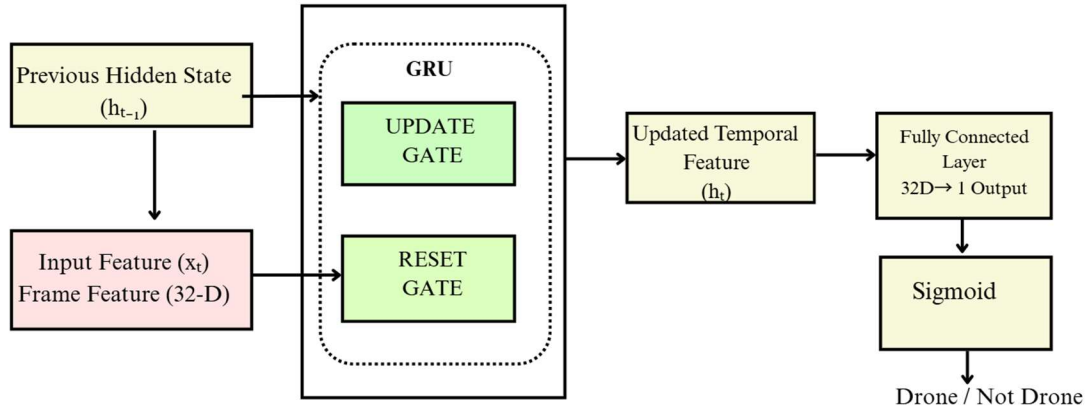


Fig 4.4 – GRU Architecture

4.4.5. Late-Fusion decision mechanism

Late fusion is based on the summation of the confidence scores of all existing modalities using a weighting scheme. Each modality makes contributions in accordance with their prediction accuracy. Hence, it is obvious that the contribution of the faulty or unreliable input will be significantly less. The obtained sum is then compared to the threshold of 0.5 to determine whether there is a drone. Those scores whose value exceeds the threshold indicate the existence of the target, and those below this indicate the absence of the target. It is important to note that such a decision-making procedure enables maintaining a high level of detection regardless of the presence or absence of one of the sensing channels due to the use of multiple modalities.

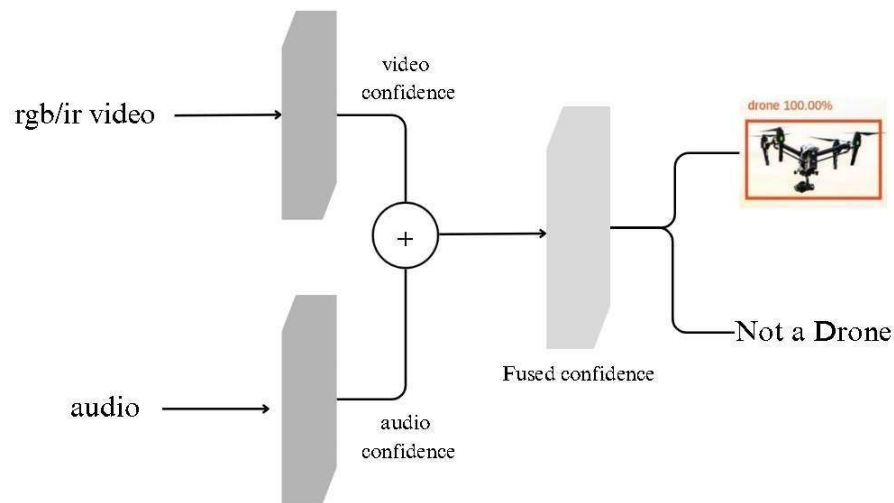


Fig 4.5- Late fusion Architecture Diagram

The Fig 4.6 demonstrates how your system will be a kind of smart referee that listens and keeps an eye at the same time and then will call upon the fact that it is a drone or not.

One of the branches analyses the RGB/IR video and gives an output of how certain a drone is or not whereas the other only analyzes the audio signal and does the same.

The two scores of confidence are then merged and form one stronger belief hence when one modality is doubtful then the other can support that and hence the final decision will be more certain.

4.5 Algorithms Used

The proposed multimodal drone detection algorithm uses various algorithms, including preprocessing, feature extraction, and decision-making algorithms that aid in the analysis of RGB videos, IR videos, and audio signals. The algorithms help in extracting relevant

acoustic, spatial, and temporal features from the raw data, and making the algorithm robust to different environmental conditions.

Motion-based frame selection, recurrent capsule networks (RCapsNet), gated recurrent units (GRU), spectrograms, and a fusion algorithm based on confidence levels are some of the algorithms used in the proposed multimodal drone detection algorithm.

4.5.1 Recurrent Capsule Network (RCapsNet)

The conventional CNN models have been efficient in identifying the spatial information of objects; nevertheless, there has been a loss of spatial information as a result of the pooling operation. For drone recognition purposes, maintaining the spatial information becomes vital since drones have unique parts that form a consistent geometric shape.

The Capsule Networks tackle this challenge through encoding the feature vectors as vectors rather than scalar quantities, whereby each vector carries spatial attributes of the object. The RCapsNet model is used in this research for spatial feature identification in drone detection applications.

Algorithm: Capsule Feature Extraction
INPUT Feature vector from convolutional layers (h)
OUTPUT Capsule-based feature representation (z)
FUNCTION: RCAPSNET_FORWARD(h)

FOR each capsule k DO

Compute prediction vector:

$$u_k = W_k h_t$$

Apply squashing function:

$$v_k = \frac{\|u_k\|}{1 + \|u_k\|^2}$$

END FOR

Aggregate capsule outputs:

$$z = \text{mean}(v_k)$$

RETURN z

END FUNCTION

In the proposed system, RCapsNet is applied to both RGB and IR modalities after preprocessing. The selected motion frames are passed through convolutional layers followed by capsule layers to extract spatial features. These features preserve structural information of drone components, which improves detection accuracy under varying orientations and backgrounds.

4.5.2 Gated Recurrent Unit (GRU)

Background and Motivation

For drone detection within video sequences, both spatial perception and temporal analysis

are essential because drones tend to display motion patterns through multiple frames. Classical feedforward models struggle to incorporate such temporal dependencies. RNNs resolve this problem but may still experience difficulties due to vanishing gradients when processing long sequences.

GRUs represent an advanced version of RNNs where gating operations regulate information flow for effective learning of temporal patterns. Compared to LSTMs, GRUs are less computationally demanding yet deliver similar accuracy in sequence prediction tasks.

Algorithm: GRU Forward Pass
INPUT Sequence of feature vectors x_t
OUTPUT Hidden state representation h_t
FUNCTION: GRU_FORWARD(x_t, h_{t-1}) Update gate: $z_t = \text{sigmoid}(W_z \times [h_{t-1}, x_t])$ Reset gate: $r_t = \text{sigmoid}(W_r \times [h_{t-1}, x_t])$ Candidate activation: $\tilde{h}_t = \tanh(W_h \times [r_t * h_{t-1}, x_t])$ Final hidden state:

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

RETURN h_t

END FUNCTION

In the proposed system, GRU is used to model temporal dependencies across sequences of frames and audio segments. For RGB and IR modalities, features extracted using RCapsNet from 40 selected frames are passed sequentially to the GRU to capture motion patterns of drones over time.

For the audio modality, GRU processes temporal features derived from spectrogram representations to capture variations in drone acoustic signatures. This temporal modeling significantly improves the system's ability to distinguish drones from other moving objects or background noise.

4.5.3 Convolutional Neural Network (CNN) for Audio Processing

Detection of drones based on acoustic analysis is based on the identification of acoustic signatures that characterize unique frequencies and harmonics produced by the rotating drone's propeller blades. Since these features are hard to describe in the wave-form format, the necessity of converting audio signal to a time-frequency domain arises.

Convolutional Neural Networks (CNN) prove to be a powerful instrument for extracting spatial features from structured data. Using log-mel spectrums as input, CNN is able to learn features corresponding to the frequency-time characteristics of the drone acoustic

signatures. Thus, CNN proves to be useful for extracting acoustic features before processing with GRU networks.

Algorithm: Feature Extraction using CNN

Within the proposed system, raw audio signals will first be converted to log-mel spectrums to encode frequency and time properties of acoustic signals. The next step will involve using CNN for extraction of high level acoustic features from log-mel spectrums. Then extracted features will be used as an input for GRU networks.

INPUT: Log-mel spectrogram

OUTPUT: Feature representation

FUNCTION CNN_FEATURE_EXTRACTION(Spectrogram)

 Apply convolution layer

 Apply activation function (ReLU)

 Apply pooling operation

 Repeat convolution and pooling layers

 Flatten feature maps

 RETURN feature vector

END FUNCTION

4.5.4 Log-Mel Spectrogram Transformation

Audio signals are challenging to comprehend due to their lack of frequency-domain properties. But drone sounds contain distinct frequency characteristics that emerge due to

rotation of propellers. For understanding frequency features in the audio signals, transformation into time-frequency domain is essential.

Mel-spectrograms have been utilized extensively due to its resemblance to human auditory sensation. Besides, taking logs of audio spectra reduces the wide span of the dynamic range in the signals.

Algorithm: Log-Mel Spectrogram Generation

Log-mel spectrograms will be generated from audio signals using pre-defined time frames, to be used in feature extraction in the proposed CNN model.

INPUT: Audio signal

OUTPUT: Log-mel spectrogram

FUNCTION LOG_MEL_SPECTROGRAM(Audio)

Load audio signal

Resample to fixed sampling rate

Apply Short-Time Fourier Transform (STFT)

Compute power spectrum

Apply mel filter bank

Convert to mel scale representation

Apply logarithmic scaling

RETURN log-mel spectrogram

END FUNCTION

By representing audio in the time–frequency domain, the system effectively captures drone-specific acoustic signatures, such as harmonic frequency patterns, which improves classification accuracy when combined with temporal modeling using GRU.

4.5.5 Motion-Based Frame Selection (RGB)

In real-world video data, not all frames contain useful information for drone detection. Many frames may consist of static background such as sky, trees, or buildings, which can introduce noise and increase computational cost. Processing all frames uniformly may reduce detection efficiency and accuracy.

To address this issue, a motion-based frame selection approach is employed. This method identifies frames containing significant motion, which are more likely to include drone activity. By focusing only on informative frames, the system reduces redundancy and improves model performance.

Algorithm: Motion-Based Frame Selection
INPUT: Sequence of RGB video frames OUTPUT: Selected motion frames
FUNCTION RGB_MOTION_SELECTION(Frames) FOR each frame t DO Convert frame to grayscale Apply contrast enhancement (CLAHE) Compute frame difference:

```


$$D_t = |\text{Frame}_t - \text{Frame}_{\{t-1\}}|$$

Apply threshold to obtain motion mask

Apply morphological operations to refine mask

IF motion detected THEN
    Mark frame as important
END IF
END FOR

Select top 40 motion frames

RETURN selected frames

END FUNCTION

```

In the proposed system, RGB videos are first converted into frame sequences. Motion-based preprocessing is then applied using frame differencing, contrast enhancement, and morphological operations to identify regions with movement.

From these, a fixed number of frames (40 frames) are selected and resized before being passed to the RCapsNet + GRU model. This ensures that the model focuses on frames containing potential drone activity, thereby improving detection accuracy and reducing unnecessary computation.

4.5.6 Infrared Motion Detection using DualMOG2

The infrared (IR) sensor technology uses thermal information, which makes it effective when there is no light or darkness at night when RGB sensors do not perform well. IR videos sometimes have noise and dynamic background changes due to environmental temperature changes.

The use of background subtraction methods helps in extracting thermal motion information in infrared videos. In this paper, the Mixture of Gaussians (MOG2) method assumes that all pixels follow a distribution of intensity and differentiates the background and foreground. This paper utilizes the DualMOG2 method by merging two methods, slow and fast background models.

Algorithm: DualMOG2 Background Subtraction
INPUT: Sequence of IR video frames OUTPUT: Motion-detected frames
<pre>FUNCTION IR_MOTION_DETECTION(Frames) Initialize slow background model Initialize fast background model FOR each frame DO Apply slow MOG2 model → FG_slow Apply fast MOG2 model → FG_fast Combine foreground masks: FG = FG_slow OR FG_fast Apply morphological operations to clean noise IF motion detected THEN Mark frame as important END IF</pre>

```
END FOR

Select top 40 motion frames

RETURN selected frames

END FUNCTION
```

In this study, the frames from an infrared video stream are fed into the DualMOG2 to extract thermal motion information. The slow model detects stable background information, and the fast model detects quick changes in the background.

This output helps in detecting moving objects like drones with minimal false alarms in the background. The detected frames are then resized and fed to the RCapsNet + GRU network.

4.5.7 Multimodal Fusion Strategy

The RGB, IR, and Audio sensing modes have distinct advantages in detecting drones. Depending on their capabilities and environment, each mode provides its unique insights into potential drone targets. Using any single sensing modality alone might result in false detections in situations like inadequate lighting conditions, obstructions, or background noises.

A fusion scheme that uses outputs from all existing sensing modes allows the system to be more resilient in cases of inaccurate estimations due to a particular sensing mode's limitations. By fusing data from multiple sensors, the system becomes capable of overcoming any issues inherent to individual sensing modes and making better decisions based on combined information inputs.

Algorithm: Multimodal Fusion

INPUT : Probabilities from RGB, IR , audio are P_{rgb} , P_{ir} , P_{audio} respectively
OUTPUT: Final decision (Drone / Not Drone)
FUNCTION MULTIMODAL_FUSION(P_{rgb}, P_{ir}, P_{audio}) Collect available probabilities Apply static weighted fusion: Assign weights to each modality Compute weighted average Compute confidence for each modality: $C_i = 2 * P_i - 0.5 $ Apply dynamic fusion: Weight each modality based on confidence Compute final probability P IF $P \geq 0.5$ THEN RETURN Drone ELSE RETURN Not Drone END IF END FUNCTION

In this proposed system, each sensing modality determines an estimation of how likely it is that there is a drone within the detection range. The probability of drone detection is then combined by applying static weighting.

In addition to this, a more sophisticated fusion algorithm is used that relies on dynamically adjusted weights depending on the confidence of each sensing modality.

This approach enables the system to perform reliably even when one or more modalities are weak or unavailable, thereby improving overall detection accuracy and robustness in real-world scenarios.

4.6 Experimentation

This chapter presents the experimental procedures carried out to evaluate the performance of the proposed intelligent multi-sensing drone detection system. The system is implemented using deep learning models for RGB, infrared (IR), and audio modalities, followed by a fusion mechanism for final decision making. The experiments focus on training, validation, and real-time inference using multimodal data.

4.6.1 Experimental Setup

These tests were conducted using a system that supports GPU, thus allowing deep learning model training. The software development process was coded in Python using PyTorch as the main deep learning software. The OpenCV framework was used as the video analysis tool for extracting frames and detecting movement, while mathematical operations were done using the NumPy software.

The log-mel spectrogram creation for audio analysis was conducted using the Librosa library. Visualization was done using the Matplotlib framework, while GUI was done using the PyQt5 library. The availability of the GPU in this platform allowed deep

learning models to train quickly since large multimodal datasets could be processed effectively.

4.6.2 Implementation Details

The suggested framework is made up of three independent models designed for processing RGB, IR, and auditory modalities. In case of RGB and IR modality inputs, pre-processing is conducted on video clips to turn them into frame sequences; then, the motion-based approach is applied for selecting informative frames.

Both RGB and IR models utilize the RCapsNet and GRU networks; here, RCapsNet extracts spatial features while GRU encodes temporal ones. As for the auditory model, the log-mel spectrogram is derived from the input signal, and both CNN and GRU models are used for further processing of this information.

All the outputs from each of these three models are finally merged to provide a final prediction result.

4.6.3 Training Configuration

All the models have been trained by employing Adam optimization method with different learning rates according to each modality. The learning rate of has been chosen for training the models for RGB and IR categories, while for the audio category, a learning rate of has been chosen.

The batch size of 4 has been employed for RGB and IR models due to high computational load, while a batch size of 16 has been utilized for the audio model. Binary Cross Entropy

loss function with class weights has been employed for the RGB and IR categories, while Cross Entropy loss with class weights has been considered for the audio category.

4.6.4 Evaluation Metrics

The performance of the proposed system was evaluated using classification accuracy and loss metrics during training and validation.

Accuracy measures the proportion of correctly classified samples, while loss indicates the error between predicted and actual labels. During inference, probability scores generated by each modality are used to compute the final decision through fusion.

4.6.5 Testing and Inference

The trained models were incorporated into a GUI on a desktop that performed real-time inference. The input data consisted of RGB video, IR video, and audio data. Preprocessing of each modality was performed, followed by inference using the corresponding model.

Probability scores are produced by each modality, which are combined using both static and dynamic approaches to fusion. The results are presented in the GUI, along with confidence scores and visual outputs like bounding boxes. The system was evaluated using various scenarios that included both drone and non-drone instances.

4.7 Design

4.7.1 Use Case Diagram

A use case diagram shows how the user interacts with the multimodal drone detection system. The first actor in this diagram is the user, who can undertake activities such as loading models for detection, uploading RGB videos, IR videos, and sound data, as well as undertaking the process of detection. The system uses the inputs provided by the user and carries out drone detection based on different models before coming up with the final fusion outputs.

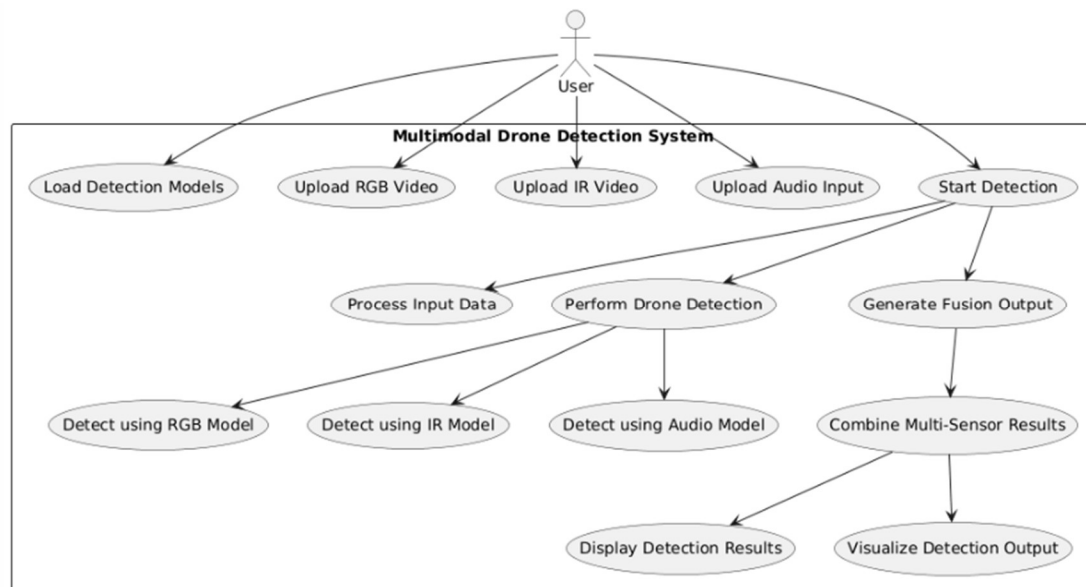


Fig 4.6- Use case Diagram

4.7.2 Class Diagram

The class diagram above shows the organizational structure of the proposed multimodal drone detection system. As you can see, various elements such as the inputs, pre-

processing, and RGB, IR, and audio-specific models work together. This allows for the processing of the inputs into probability outputs that are later fused together using an appropriate method. The decision is made using the output from the fusion process and presented to the user using the visualization component.

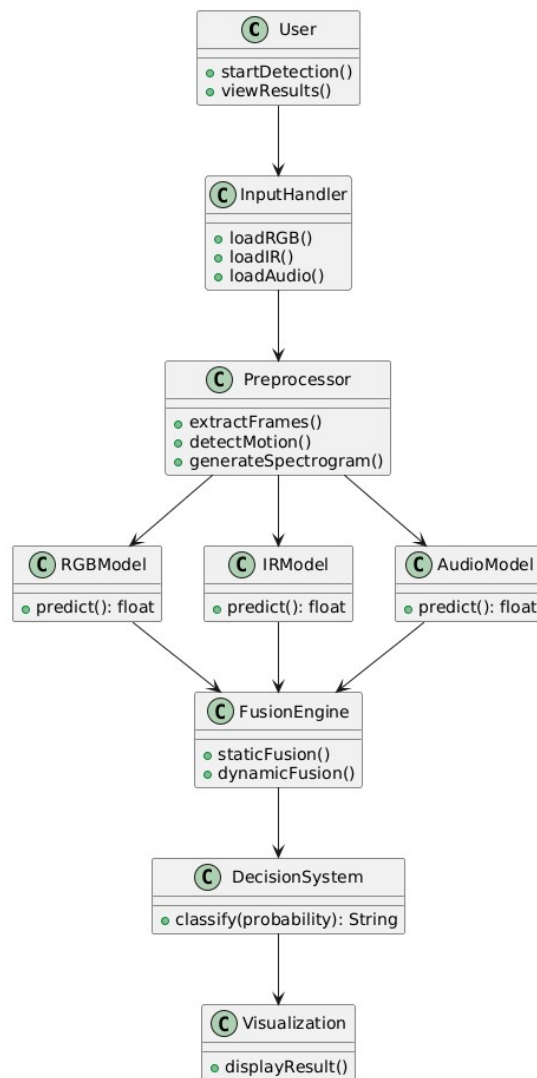


Fig 4.7- Class Diagram

4.7.3 Sequence Diagram

The sequence diagram represents the interaction between different components of the system during the drone detection process. It shows how the user initiates the detection through the interface, after which the system performs preprocessing tasks such as frame extraction, motion detection, and spectrogram generation. The processed data is then forwarded to the respective RGB, IR, and audio pipelines for feature extraction and temporal modeling. The generated probability outputs are sent to the fusion module, where confidence-based weighting is applied to compute the final decision. The result is then returned to the user interface for display.

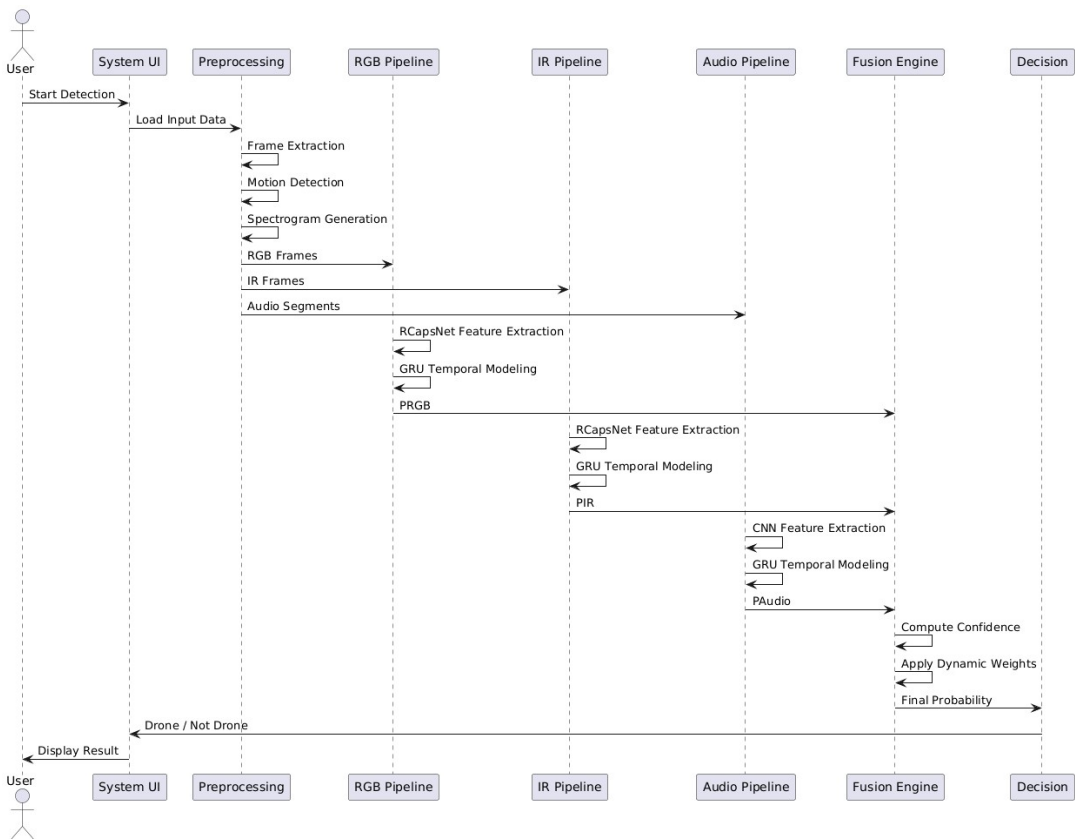


Fig 4.8 Sequence Diagram

4.7.4 Activity diagram

The activity diagram illustrates the overall workflow of the proposed multimodal drone detection system. It begins with capturing RGB, infrared (IR), and audio inputs, followed by preprocessing and motion-based frame selection. The processed data is then passed through three parallel pipelines corresponding to RGB, IR, and audio modalities, where feature extraction and temporal modeling are performed using RCapsNet, CNN, and GRU architectures. The outputs from each modality are converted into probability scores, which are further combined using static or dynamic fusion strategies. Based on the final probability, the system classifies the input as drone or non-drone and displays the result.

4.7.5 State diagram

The state diagram represents the various states and transitions in the multimodal drone detection process. The process starts in the idle state when the system is idle and awaiting input from the user. Once the detection process is initiated, the process moves to the data loading and preprocessing stage. After that, the system enters the processing stage where the RGB, IR, and audio features are processed in parallel. The probability is computed using the features in each modality, and the results are fused together to make a decision. Based on the results obtained, the system moves to either the drone detected or not drone state before finally moving back to the idle state.

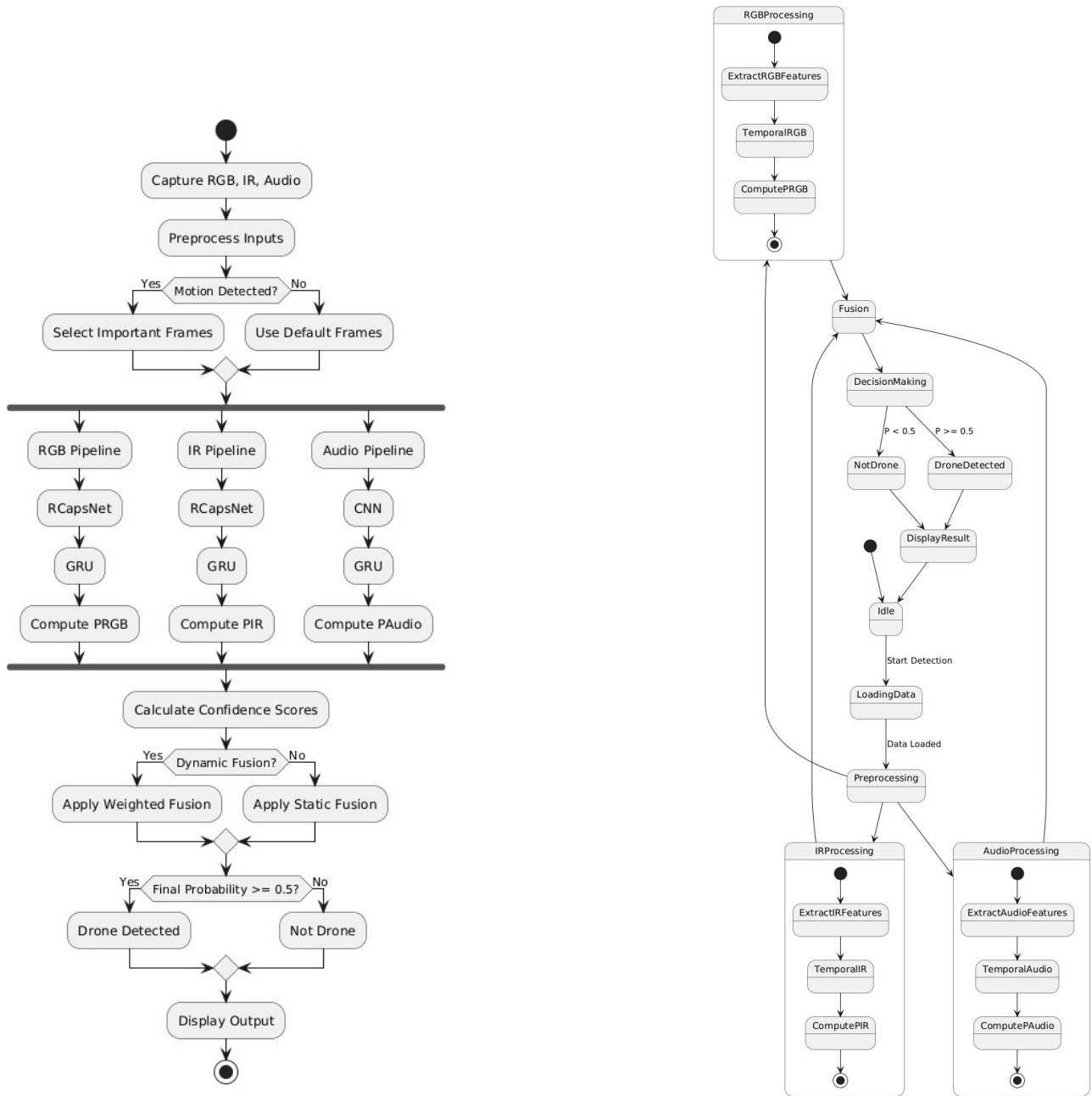


Fig 4.9 - Activity and State Diagram

CHAPTER 5

RESULTS

In this section, a large-scaled study of this multimodal drone detection system in 14 specially designed inference scenarios is proposed. The test sample is designed to go through three increasing steps: single-modality assessment (Group A), dual-modality assessment (Group B), and tri-modality assessment (Group C). Sensing strategies are not only quantitatively compared across configurations, but also using both static and dynamic fusion scores and per-modality probability outputs. The summarized data are presented in Table 1, which is the main source of data in this discussion.

Table 5. 1 - Quantitative Evaluation of the Proposed Multimodal Drone Detection System

Scenario	Input Modality	Ground truth	P_rgb	P_ir	P_audio	P_static	P_Fusion	Decision
ID	Modalities used	Label	RCapsGR U_RGB	RCapsGRU _IR	Audio_C NN_GRU	W = 0.4/0.35.0.2 5	$ p-0.5 ^2$	Drone/Not Drone
Group A – Single Modality (6 Scenarios)								
S1	RGB Only	Drone	0.912	N/A	N/A	0.912	0.912	Drone
S2	RGB Only	NotDrone	0.032	N/A	N/A	0.032	0.032	Not Drone
S3	IR Only	Drone	N/A	0.928	N/A	0.928	0.928	Drone
S4	IR Only	Not Drone	N/A	0.048	N/A	0.048	0.048	Not Drone
S5	Audio Only	Drone	N/A	N/A	0.996	0.996	0.996	Drone
S6	Audio Only	Not Drone	N/A	N/A	0.01	0.01	0.01	Not Drone
Group B – Dual Modality (6 Scenarios)								
D1	RGB + IR	Drone	0.907	0.961	N/A	0.933	0.936	Drone

D2	RGB + IR	NotDrone	0.035	0.297	N/A	0.157	0.115	Not Drone
D3	RGB + Audio	Drone	0.904	N/A	0.290	0.668	0.694	Drone
D4	RGB + Audio	Not Drone	0.085	N/A	0.030	0.053	0.040	Not Drone
D5	IR + Audio	Drone	N/A	0.950	0.743	0.864	0.878	Drone
D6	IR + Audio	Not Drone	N/A	0.051	0.010	0.030	0.0250	Not Drone
Group B – Full Tri Modality (2 Scenarios)								
T1	RGB + IR + Audio	Drone	0.903	0.935	0.68	0.858	0.877	Drone
T2	RGB + IR + Audio	Not Drone	0.913	0.051	0.01	0.383	0.294	Not Drone

5.1 Single-Modality Baseline Performance

The unimodal experiments (S1-S6) determine the independent detection of the RGB, the infrared, and the audio sensing branches before fusion. These baseline experiments confirm that all senses have the ability to differentiate between drone and non-drone aerial targets with unimodal information in a correct way.

5.1.1 RGB Modality

In Scenario S1, the RGB-based detection model (RCapsNet + GRU) has a detected drone probability of $P_{RGB} = 0.912$ as in Fig: 5.1, which illustrates that spatial and temporal information contained in the visible video stream are very discriminating when the visualization is good.



Fig – 5.1 - RGB-only drone detection ($P_{\text{RGB}} = 0.912$). The RCapsNet--GRU model correctly identifies the drone from the visible video stream. Green bounding box indicates the localized target.

At the non-drone Scenario S2, the RGB model generates a low confidence value of $P_{\text{RGB}} = 0.032$, which means that there is a high rejection of the visually different aerial objects. This validates that drone-specific visual patterns are well encoded in capsule-based spatial encoding in presence of background clutters. Nevertheless, later in Scenario T2 it can be noticed that the RGB branch is vulnerable to visual ambiguity in case non drone objects exhibit drone-like visual properties.

5.1.2 Infrared Modality

In Scenario S3, the infrared detection model has a confidence score of $P_{\text{IR}} = 0.928$, which is a bit better than RGB. Scenario S3 and utility of thermal imaging in monitoring the heat signatures of drone motors and propellers are shown in Fig: 5.2.



Fig – 5.2 - IR-only drone detection ($P_{IR} = 0.928$). The thermal heat signature of the drone is reliably captured by the RCapsNet--GRU model operating on single-channel infrared frames.

In Scenario S4 (non-drone input) the IR branch rightly gives low confidence score of $P_{IR} = 0.048$. Obviously, this is a significant benefit of thermal imagery over the more obscure RGB response in Scenario T2, where the drone can nearby non-drones exhibit significant heat concentrations that are often well-focused on the propulsion such as in the case of a multi-rotor drones.

5.1.3 Audio Modality

The audio-based detection model has the largest misclassification confidence in comparison to all other branches of unimodal sensing. In Scenario S5, system obtains a drone detection confidence score of 0.996 as shown in Figure 5.3, where the log-mel spectrogram distinctly shows the harmonic structure caused by the rotation of the drone

propellers. This verifies that, acoustic sensing offers very unique frequency domain elements to identify drones..

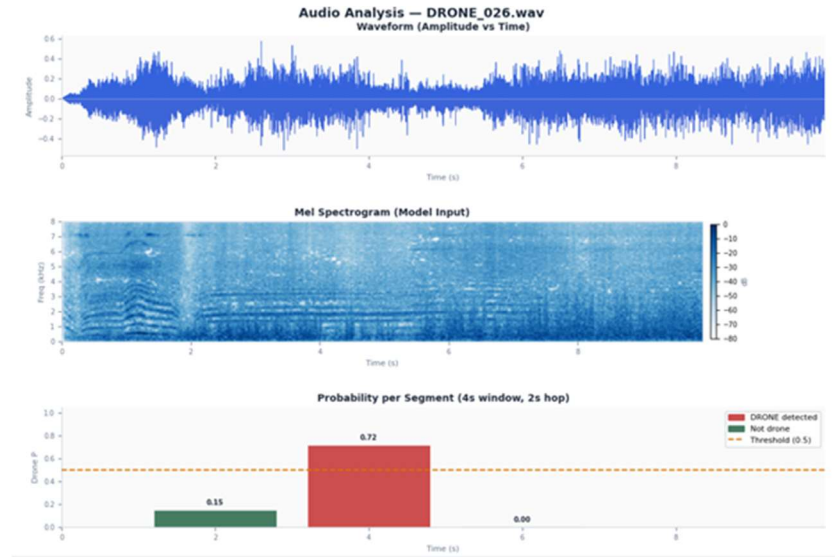


Fig – 5.3 - Audio-only drone detection ($P_{\text{AUDIO}} = 0.996\%$). The log-mel spectrogram captures the characteristic harmonic structure of drone propeller rotation.

In Scenario S6 the audio sensing branch accepts the non-drone aerial object with a confidence score of 0.010 and is the highest of all unimodal detection branch separations between the drone and non-drone confidence scores. The high discrimination ability is caused by the acoustic signatures that are generated by multi-rotor propulsion systems.

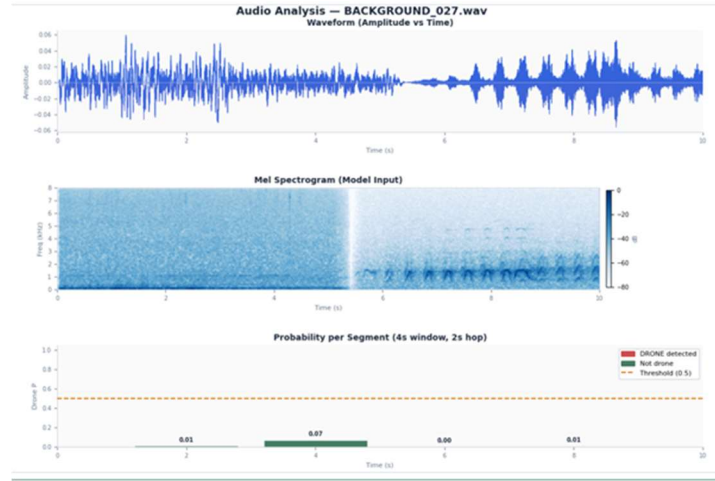


Fig – 5.4 - Audio non-drone rejection ($P_{\text{AUDIO}} = 0.010$). The log-mel spectrogram captures the characteristic harmonic structure of drone propeller rotation, enabling near-certain classification.

5.2 Dual-Modality Fusion Performance

Once the engineering of individual sensing has been efficient, fusion devices with dual-modality assess the enhancement of detection reliability and the decrease of risk of false activation through the combination of two sensing channels.

In D1, the combination of the RGB and infrared sensors yields a drone detection confidence of 0.933 with the static fusion and 0.936 with the dynamic fusion. This shows that a mixture of thermal and visible-spectrum information yields higher confidence of detection as compared to single analysis. This enhancement indicates that both complementary spatial and thermal characteristics can be used with each other to promote reliability in detection..



Fig – 5.5 – RGB+IR dual-modality fusion ($P_{\text{dynamic}} = 0.936$). The combination of visible and thermal evidence produces a higher fused confidence than either modality individually, demonstrating the additive benefit of complementary visual sensing.

Scenario D2 contains the correct fusion configuration where the non-drone object is correctly classified with the confidence value of 0.157 (static fusion) and 0.115 (dynamic fusion). The smaller dynamic fusion score is an indicator that the confidence-weighted fusion mechanism is effective at diminishing the roles of less dependable modality responses, causing an increase in classification certainty as shown in Fig 5.5. Further evidence of complementary sensing behavior is attained in the RGB and audio fusion setup tested in Scenario D3.

Even though the audio alone news item has a moderate score of confidence of 0.290, the good RGB response of 0.904 prevails in the fusion response and results in a correct classification of the drone. This verifies that audio knowledge has assistive reinforcement in a combination with powerful visual evidence. In Scenario D4, the same fusion setup rejects the non-drone input, with the dynamic fusion confidence value of 0.040, which means that the suppression of false activation is achieved via multimodal agreement. The

best dual-modality is in Scenario D5, in which audio and infrared sensing are implemented. With this setup, the system generates fused scores of confidence of 0.864 (static fusion) and 0.878 (dynamic fusion). This finding illustrates that thermal spatial distribution and rotor acoustic signatures give very complementary detection evidences. The equals layout in Scenario D6 generates a very low confidence of 0.025 in the non-drone case, which is a countercheck of a power to reject..

5.3 Full Tri-Modal Fusion Performance

The overall suggested system will integrate RGB, infrared and audio sensing in real time to generate the complete classification choice.

Scenario T1 has the individual modalities confidence scores of 0.903 (RGB), 0.935 (infrared), and 0.680 (audio). Combining these two results in confidence values of 0.858 when fused in a static way and 0.877 when fused in a dynamic way, shown in Figure 5.7. This finding validates that the combination of three separate sensing channels generates strong detection abilities that are backed by complementary spatial, thermal, and acoustic indicators. The table below Figure 5.7 shows the results of tri-modal fusion, in which all three sensory modalities (RGB, IR and Audio) can be be included in a consistent and cooperative way to reach a final answer that is close to the ground-truth choice.

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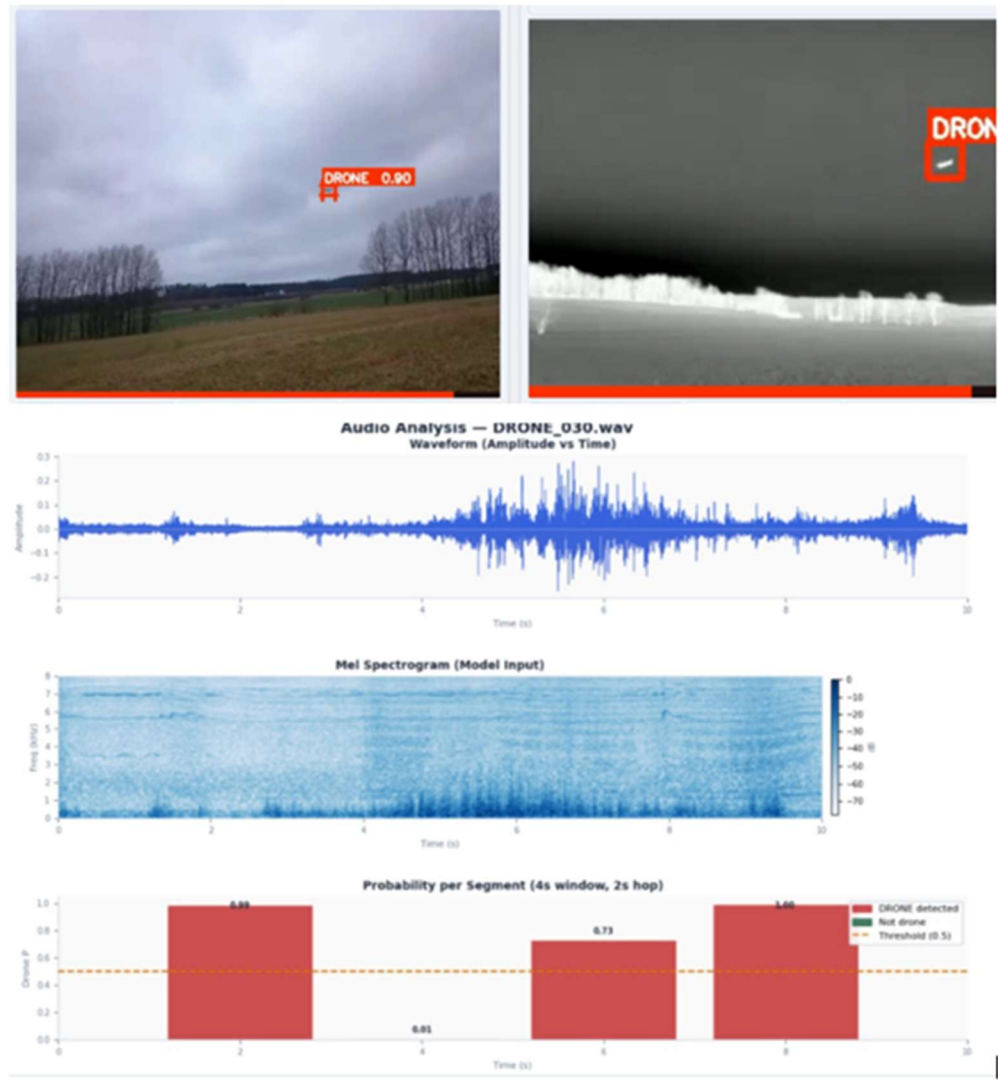


Fig – 5.6 - The complete proposed system (T1 and T2) is the tri-modal fusion scenarios, that is, the combination of all three modalities of sensing in parallel. The system output of scenario T1 (drones detection)

Scenario T2 offers a significant validation instance to understand the strength of the proposed fusion architecture in case of disagreement in modality. The RGB branch shows in this case, a falsely high confidence value of 0.913 which by itself can result in a false detection of drones. But the infrared and audio branches have low confidence scores of

0.051 and 0.010 respectively. The fusion mechanism as shown in Figure T2 manages to eliminate the undesired RGB contribution, and finally yields final confidence scores of 0.383 (static fusion) and 0.294 (dynamic fusion), which generate correct non-drone classification. The benefit of adaptive confidence-weighted fusion in avoiding false activation is clearly shown in this experiment.

5.4 Classification Performance and Confusion Matrix

The performance of the proposed multimodal detection system in general can be summarized by the overlay of the confusion matrix presented in Figure 5.8. The obtained evaluation results on 480 samples have 187 true positives, 281 true negatives, 1 false positive and 2 false negatives.

		Predicted Class	
		Positive	Negative
Actual Class	Positive	187	2
	Negative	10	281
		Actual Class	Predicted Class

Fig – 5.7 – Confusion Matrix

These findings allow concluding that the accuracy of overall classification is 97.5% which proves high reliability of the offered detection framework. The accuracy measure of 94.9% shows that the majority of observed drone events represent actual drone targets whereas the recall measure of 98.9% serves to confirm that the system is able to locate

the vast majority of any drone events. The specificity value of 96.6 per cent also indicates a high rejection ability with non-drone aerial objects. False-positive rate is minimal (3.4), which is especially significant in the real-world scenario of surveillance when actually unnecessary alarms are to be reduced to minimal. Correspondingly, the false-negative of 1.1 per cent not only proves that the number of drone events missed during detection is very small. Additional evidence of the balanced performance between detection sensitivity and false-alarm rejection is F1-score based on the accuracy between prediction reliability and classification-accuracy which is 96.8 percent and a negative predictive value amounting to 99.3 percent .

5.5 Static vs Dynamic Fusion Comparison

A comparative analysis of the effects of a static fusion strategy and dynamic fusion strategy reveals that the effect of dynamic fusion is always the ability to generate a higher level of reliability in the decision made under all the fourteen scenarios, The below Fig: 5.9 illustrates this ideal pattern of dynamic fusion.

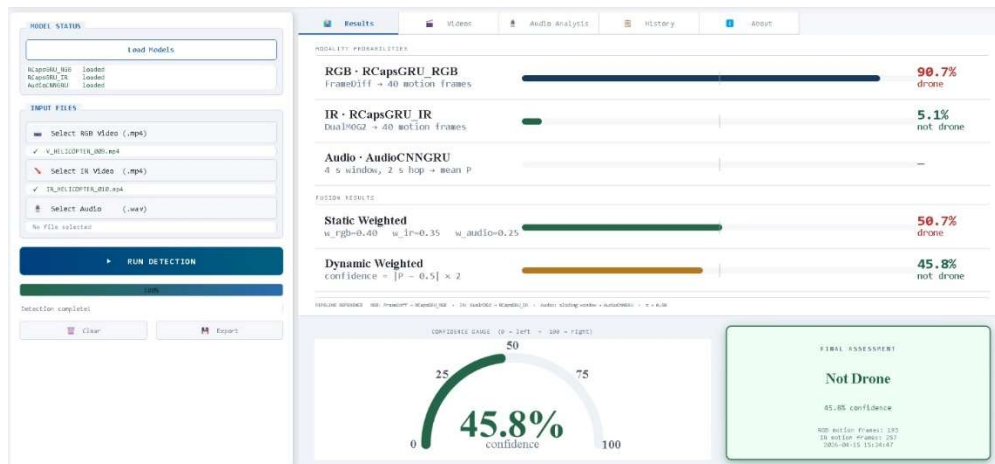


Fig 5.8 – GUI showing importance of Static vs Dynamic Fusion

To illustrate this, Scenario T1 has a confidence score of 0.877 due to dynamic fusion compared to the 0.858 of the static fusion. On the same note, under the Scenario D1, dynamic fusion generates 0.936 relative to 0.933 generated by the use of the static fusion. In the case of disagreement scenarios as in Scenario D2 and Scenario T2, dynamic fusion yields low confidence scores than those in the case of static fusion thereby generating more classification margin beyond the decision threshold and minimizing the risk of false-positive. This pattern indicates that the weighted fusion mechanism based on confidence is effectively used in order to repress unsound modality reply on inference.

5.6 Summary of Results

Three significant findings are confirmed in the course of the experimental evaluation. First, RGB, infrared, and audio sensing modalities have complementary information on detection in the spatial, thermal, and acoustic space. Second, multimodal fusion always yields high detection reliability than that of single-modality performance, especially in infrared and audio fusion systems. Third, dynamic confidence-weighted fusion is much robust than the static fusion because it is misleading modality responses when there exists disagreement. On balance, it can be concluded that the suggested tri-modal fusion solution is a stable and effective approach to intelligent drone surveillance under complex conditions in the real world.

CHAPTER 6

CONCLUSION

This project demonstrated a multimodal drone detection system, which incorporates both visual (RGB) and infrared (IR) and audio sensory data in a cohesive deep learning system. This scheme is motivated by the weaknesses of single-modality detection schemes that tend to do poorly in adverse real-world scenarios like changes in illumination, acoustic crowding, and background clutters. The proposed system enhances the robustness of detection in a variety of surveillance settings by integrating complementary senses.

The system uses Recurrent Capsule Neural Network (RCapsNet) with GRU layers to process RGB and IR video images and CNN-GRU pipeline to achieve audio-based detection, which is based on log-mel spectrograms. These branches have their outputs fused with the combination of both the static and dynamic fusion schemes, which allow the at will weighting of trusted modalities during inference. Simulated testing in fourteen structured scenarios has shown excellent drone detection and good non-drone object rejection, and dynamic fusion can effectively address modality disagreement situations like Scenario T2.

The framework was also used to process desktop-based inference application which showed that it could be used practically in this UAV surveillance application. Altogether, the findings indicate that the presented multimodal architecture offers enhanced (i.e., a higher) detection accuracy, robustness, and reliability over single-sensors.

CHAPTER 7

FUTURE SCOPE

Although promising results have been demonstrated in the proposed system, there are some factors that can restrict its application. Improvements are needed.

The existing system works by processing modality individually followed by fusing. Future studies may consider attention-based cross-modal fusion methods or transformer-based models. These would permit sharing more information about features across modalities earlier, possibly at a higher level of detail of sensor-sensor relationships.

Second, the existing appraisal is based on the data that is gathered under certain conditions outside. The applicability of the trained models could be improved by expanding the system to include other datasets to increase the diversity of the screened models; this includes: urban environment, altitude variability, drone aggressive behaviors, and the drone itself.

Third, the existing audio model is based on fixed length segments. It could help enhance real-time responsiveness and decrease latency by implementing event-based or adaptive audio processing. Lastly, the system should be equipped with drone localization and trajectory prediction to make it much more useful during the active counter-UAV missions since determining where a detected drone is located is essential to responding to it.

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